NOBEL PRIZES

DATA SCIENCE REPORT

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Nobel Prize History Report

The Nobel Prize is considered one of the most prestigious awards for human achievement and honors individuals and organizations that have made significant contributions to science, literature, peace, and economics. Since its establishment in 1895 at the behest of Alfred Nobel, the Prize has honored those whose work has shaped the course of human history. Nobel's vision was clear: he wanted to recognize those who had "rendered the greatest service to mankind." Over the course of more than a century, this vision has evolved into a global symbol of excellence, innovation, and effectiveness.

In this report, we use data science methods to examine hidden within the history of Nobel laureates between 1901 and 2020. Using tools such as Python, Pandas, and interactive visualizations, we analyze trends across countries, cities, institutions, and price categories. We are going to go beyond simply identifying the winners and explore deeper questions: Which countries dominate? How has the age of the winners changed over time? Do certain institutions consistently produce winners? By examining these aspects, this analysis aims to draw meaningful conclusions and provide a more comprehensive perspective on how excellence, recognition, and global progress are interconnected through the Nobel Prize.

Source Data Files (CSV Format)

Nobel Prize Data: github.com

Import lines:

```
import pandas as pd
import numpy as np
import plotly.express as px
import seaborn as sns
import matplotlib.pyplot as plt
import plotly.graph_objects as go
import statsmodels
```

Display Formatting for Financial Data

To improve readability and ensure consistent formatting, we are going to configure pandas to display floating-point numbers and values in a more appropriate way.

```
# Notebook Presentation:
pd.options.display.float_format = '{:,.2f}'.format
pd.set_option('display.max_columns', None)
pd.set_option('display.width', None)
```

Acquiring the Source and Exploring Initial Dataset

```
# Retrieving Data
df_data = pd.read_csv('csv_files/nobel_prize_data.csv')
```

We can see that the data covers the Nobel laureates from 1901 to 2020.

```
# Exploring the Initial data

print("Shape of the dataframe:")
print(df_data.shape)

print("\nFirst five entries")
print(df_data.head(5))
```

Shape of the DataFrame:

```
(962, 16)
```

The findings reveal that we have 962 rows (entries) and 16 columns (features) of Nobel laureates.

First five Entries:

Given the size of the dataset, we will display these five entries in three sections.

	year	category	prize	motivation
0	1901	Chemistry	The Nobel Prize in Chemistry 1901	"in recognition of the extraordinary services ...
1	1901	Literature	The Nobel Prize in Literature 1901	"in special recognition of his poetic composit...
2	1901	Medicine	The Nobel Prize in Physiology or Medicine 1901	"for his work on serum therapy, especially its...
3	1901	Peace	The Nobel Peace Prize 1901	NaN
4	1901	Peace	The Nobel Peace Prize 1901	NaN

	prize_share	laureate_type	full_name	birth_date	birth_city	birth_country \
0	1/1	Individual	Jacobus Henricus van 't Hoff	1852-08-30	Rotterdam	Netherlands
1	1/1	Individual	Sully Prudhomme	1839-03-16	Paris	France
2	1/1	Individual	Emil Adolf von Behring	1854-03-15	Hansdorf (Lawice)	Prussia (Poland)
3	1/2	Individual	Frédéric Passy	1822-05-20	Paris	France
4	1/2	Individual	Jean Henry Dunant	1828-05-08	Geneva	Switzerland

	birth_country_current	sex	organization_name	organization_city	organization_country	ISO
0	Netherlands	Male	Berlin University	Berlin	Germany	NLD
1	France	Male	NaN	NaN	NaN	FRA
2	Poland	Male	Marburg University	Marburg	Germany	POL
3	France	Male	NaN	NaN	NaN	FRA
4	Switzerland	Male	NaN	NaN	NaN	CHE

These early entries show that the Nobel Prize was established in 1901 with a strong European focus, as all of the early laureates were born in countries such as France, the Netherlands, Switzerland, and regions of historic Prussia. Furthermore, certain patterns are already obvious throughout the dataset, such as the dominance of individual laureates, the existence of jointly awarded prizes (such as in the “Peace” category), and the close connection between academic institutions and scientific awards.

Understanding the Dataset Structure

Before we proceed to a more in-depth analysis, it is essential to understand the structure and quality of the dataset. Using the .info() function in Python, we can obtain a clear summary of the DataFrame, including the number of entries, column names, data types, and non-zero values. This step provides a quick overview of the available information and helps identify missing or inconsistent data that could affect the analysis.

```python
print("\nOverview of the Dataframe")
print(df_data.info())
```

```
RangeIndex: 962 entries, 0 to 961
Data columns (total 16 columns):
 #   Column                 Non-Null Count  Dtype
---  ------                 --------------  -----
 0   year                   962 non-null    int64
 1   category               962 non-null    str
 2   prize                  962 non-null    str
 3   motivation             874 non-null    str
 4   prize_share            962 non-null    str
 5   laureate_type          962 non-null    str
 6   full_name              962 non-null    str
 7   birth_date             934 non-null    str
 8   birth_city             931 non-null    str
 9   birth_country          934 non-null    str
 10  birth_country_current  934 non-null    str
 11  sex                    934 non-null    str
 12  organization_name      707 non-null    str
 13  organization_city      707 non-null    str
 14  organization_country   708 non-null    str
 15  ISO                    934 non-null    str
```

The dataset contains 962 entries across 16 columns and covers various aspects of Nobel laureates, including personal details, award categories, institutional affiliations and information about research centers. While most columns are complete, some—such as those regarding motivations, fields of study at the time of birth, and information on organizations—contain missing values and require elaborate inspection.

Overall, the dataset appears comprehensive and well-structured; however, these missing values suggest that certain analyses (particularly those relating to institutions or demographic data) may require careful processing or data cleaning to ensure reliable findings.

Examining Missing and Duplicate Values

A crucial step in any data analysis is assessing the quality of the dataset, particularly by identifying missing and duplicate values. In this section, we'll use built-in Pandas functions to check whether the dataset contains missing values (NaN) or duplicate rows.

This helps ensure the reliability of our analysis and identifies areas where data cleaning or preprocessing may be necessary.

```
# Dealing with NaN and duplicate values:
print("\nIs there any NaN values?")
print(df_data.isna().values.any())
print("\nIs there any duplicated values?")
print(df_data.duplicated().values.any())
```

```
Is there any NaN values?
True

Is there any duplicated values?
False
```

The dataset contains no duplicate entries, which suggests that each Nobel Prize entry is uniquely recorded—an important advantage for accurate analysis.

However, we confirm here again that the DataFrame contains some missing values.

Looking into missing values in all Features of the DataFrame

To better understand the distribution of missing values, we are going to calculate the number of NaN values in each column.

By this way, we are going to identify exactly which variables are incomplete and assess how these gaps might affect further analysis or visualization.

```
# We can get a count of the NaN values per column using this:
print("\nNumbers of Missing Values per Column:")
print(df_data.isna().sum())
```

Numbers of Missing Values per Column:

```
year                 0
category             0
prize                0
motivation          88
prize_share          0
laureate_type        0
full_name            0
```

```
birth_date                 28
birth_city                 31
birth_country              28
birth_country_current      28
sex                        28
organization_name         255
organization_city         255
organization_country      254
ISO                        28
```

The missing values follow a clear pattern: For several fields containing personal information—such as date of birth, country of birth, and gender—approximately 28 entries are consistently missing, suggesting gaps in the historical records for a small subset of the award recipients.

In contrast, the significantly higher number of missing values in the columns for organizations suggests that many laureates, particularly in categories such as “Peace” or “Literature,” may not officially be affiliated with any institution, a factor that should be taken into account.

Insights into missing data

Now that we have determined that the Dataset contains missing values, we will examine them more closely to understand the underlying causes.

By this way, we will focus in particular on entries where personal data—such as dates of births are missing, to determine whether this is due to incomplete records or to specific characteristics of certain award recipients.

Targeted Display Filtering

Given the large number of variables in the dataset, we would like to display the values which are the most relevant particular tasks. In Pandas it requires just a simple list of desirable columns to attach the method later on.

```
# Filtering for the Display
columns_to_display = ['year', 'category', 'full_name', 'birth_date',
'birth_country', 'laureate_type','organization_name']
```

Now we are going to filter the DataFrame by passing on isna() function on birthdate column which is going to allocate the entries for missing values at this column.

```
# look what are the other colums when birth date is na:
df_no_birthday = df_data.loc[df_data.birth_date.isna()]
print(df_no_birthday[columns_to_display].head())
```

Records with missing values in the birth date field:

year	category	full_name	birth_date	birth_country	laureate_type	organization_name
1904	Peace	Institut de droit international (Institut...	NaT	NaN	Organization	NaN
1910	Peace	Bureau international permanent de la Paix...	NaT	NaN	Organization	NaN
1917	Peace	Comité international de la Croix Rouge	NaT	NaN	Organization	NaN
1938	Peace	Office international Nansen pour les Réf ...	NaT	NaN	Organization	NaN
1944	Peace	Comité international de la Croix Rouge	NaT	NaN	Organization	NaN

The filtered results show that many entries with missing birth dates are more likely to be attributed to organizations rather than individual laureates, such as committees and international institutions.

Interestingly, these organizational laureates appear predominantly in the “Peace” category, highlighting the importance of collaboration to promote peace and global cooperation.

This pattern suggests that missing personal data is not a data quality issue but rather reflects the diverse types of Nobel laureates included in the dataset.

Expanding the DataFrame by adding additional features

We are going to create two additional features derived from the existing dataset.

Deriving Prize Share Percentages

To improve our analysis, we convert the “prize_share” column into a more usable numerical format by calculating each winner’s share as a percentage.

To do this, we split the fractional values (e.g., “1/2,” “1/3”) and convert them to floating-point numbers, which allows for clearer comparisons and a quantitative analysis of the prize distribution among the winners.

```
share_of_prize = df_data.prize_share.str.split('/', expand=True)
numerator = pd.to_numeric(share_of_prize[0])
denominator = pd.to_numeric(share_of_prize[1])
df_data['share_pct'] = numerator / denominator
```

Calculating Laureates’ Winning Age

Another key characteristic we are examining is the age of individual laureates at the time the Nobel Prize was awarded.

By extracting the year of birth from the “birth_date” column and comparing it with the year the prize was awarded, we gain valuable insights into when individuals typically make their most significant contributions.

```
birth_years = df_data.birth_date.dt.year
df_data['winning_age'] = df_data.year - birth_years
```

Statistical Overview of Nobel Laureates’ Winning Ages

To better understand the age distribution of Nobel laureates, we are going to perform a descriptive statistical analysis of their ages when they received their awards.

Using the .describe() function, we calculate key summary statistics such as the mean, median, minimum, maximum, and quartiles, which provide a comprehensive overview of the age distribution of the laureates.

```
print("\nDescriptive Analysis of the Laureates’ Winning Ages:")
print(df_data.winning_age.describe())
```

Descriptive Analysis of the Laureates’ Winning Ages:

```
count   934.00
mean     59.95
std      12.62
min      17.00
25%      51.00
50%      60.00
75%      69.00
max      97.00
Name: winning_age, dtype: float64
```

The results show that the average age of Nobel laureates is around 60, with most laureates being between 51 and 69 years old, suggesting that significant, recognized achievements are often made later in life.

However, the wide age range—from 17 to 97—makes it clear that groundbreaking contributions can be made at almost any stage of life, reflecting the diversity of outstanding achievements across various fields and under different circumstances.

Identifying Outstanding Entries in the Dataset

To better understand the diversity of Nobel laureates, we can examine the two extremes of the age spectrum—those who received the prize at a remarkably young age or at an advanced age.

By filtering these exceptional cases from the dataset, we gain a more personal and human perspective on achievements at different stages of life.

```
print("\nThe Youngest Laureate:")
print(df_data.nsmallest(n=1, columns='winning_age').iloc[0])

print("\nThe Oldest Laureate:")
print(df_data.nlargest(n=1, columns='winning_age').iloc[0])
```

The Youngest Laureate:

```
year                                                                 2014
category                                                            Peace
prize                                             The Nobel Peace Prize 2014
motivation               "for their struggle against the suppression of...
prize_share                                                           1/2
laureate_type                                                  Individual
full_name                                                 Malala Yousafzai
birth_date                                             1997-07-12 00:00:00
birth_city                                                        Mingora
birth_country                                                    Pakistan
birth_country_current                                            Pakistan
sex                                                                Female
organization_name                                                     NaN
organization_city                                                     NaN
organization_country                                                  NaN
ISO                                                                   PAK
```

```
share_pct                                                                   0.50
winning_age                                                                17.00
Name: 885, dtype: object
```

As illustrated above, Malala Yousafzai became the youngest Nobel laureate in history in 2014 at the age of just 17, receiving the Peace Prize for her advocacy for girls' education. Born in Pakistan, Yousafzai gained worldwide fame after surviving an assassination attempt by the Taliban, which only further strengthened her resolve to advocate for the right to education.

Her recognition at such a young age highlights how the Nobel Prize is not limited to lifetime achievements but can also honor early, high-impact contributions to global issues. Malala's case is particularly striking because it reflects the power of youth activism and how social influence is no longer bound by age or traditional career timelines.

The Youngest Laureate:

```
year                                                                   2019
category                                                          Chemistry
prize                                          The Nobel Prize in Chemistry 2019
motivation                  “for the development of lithium-ion batteries”
prize_share                                                             1/3
laureate_type                                                    Individual
full_name                                                   John Goodenough
birth_date                                              1922-07-25 00:00:00
birth_city                                                             Jena
birth_country                                                       Germany
birth_country_current                                               Germany
sex                                                                    Male
organization_name                                      University of Texas
organization_city                                                 Austin TX
organization_country                                United States of America
ISO                                                                     DEU
share_pct                                                              0.33
winning_age                                                           97.00
Name: 937, dtype: object
```

As reflected in the data, at the age of 97, John B. Goodenough became the oldest Nobel laureate when he was awarded the 2019 Nobel Prize in Chemistry for his contribution to the development of lithium-ion batteries—a technology that powers modern electronic devices ranging from smartphones to electric vehicles.

Goodenough's achievements demonstrate that groundbreaking contributions can still be made even in old age, thereby underscoring the enduring nature of intellectual creativity and scientific relevance. His work also shows that the Nobel Prize often recognizes innovations that bring about profound change over the long term and sometimes only decades after the original discovery.

Analysis of the Gender Distribution

An Insight into the gender distribution of Nobel laureates' sheds light on representation and historical inequalities in globally recognized achievements. By analyzing this aspect, we can observe how the participation and recognition of genders have evolved over time.

This section aims to highlight both inequalities and gradual progress and to provide a data-driven perspective on inclusivity in the history of the Nobel Prize.

Forming the Dataset

To analyze the gender distribution, we group the dataset by "sex" and count the total number of prizes awarded to each group.

This aggregation allows us to transform the raw data into a clear overview, making it easier to compare gender representation.

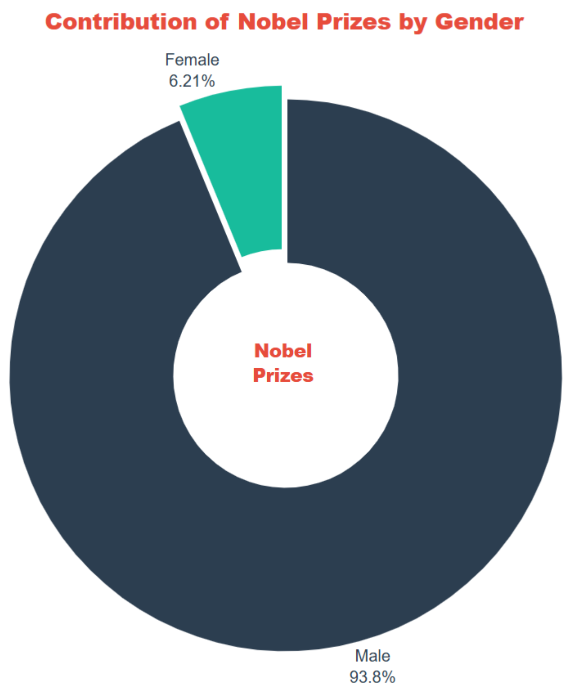
```
df_gender = df_data.groupby(["sex"], as_index=False).agg({'prize':
pd.Series.count})
```

Creating pie chart:

code:

```
pie_graph_colors = ['#18BC9C', '#2C3E50', '#E74C3C']

fig = go.Figure(data=[go.Pie(
    labels=df_gender['sex'],
    values=df_gender['prize'],
```

```
    hole=0.4,
    textinfo='label+percent',
    textposition='outside',
    textfont=dict(size=16, color='#2C3E50', family='Arial, sans-serif'),
    insidetextorientation='radial',
    marker=dict(colors=pie_graph_colors, line=dict(color='#FFFFFF', width=3)),
    pull=[0, 0.05]     # it will take a slice out.
)])

fig.update_layout(
    title=dict(text="Contribution of Nobel Prizes by Gender",
        x=0.5, xanchor='center',
        font=dict(size=22, family="Arial Black", color="#E74C3C")),
    annotations=[
        dict(
            text='Nobel<br>Prizes',
            x=0.5, y=0.5,
            font_size=18, font=dict(family="Arial Black", color="#E74C3C"),
            showarrow=False
        )
    ],
    showlegend=False,
    plot_bgcolor='white',
    paper_bgcolor='white',
    margin=dict(t=80, b=40, l=40, r=40)
)

# fig.write_html("interactive_graphs/nobel_prizes/gender_distribution.html")

fig.show()
```



Interactive Graph Link: Contribution of Nobel Prizes by Gender

The graph clearly shows a significant imbalance in the distribution of Nobel Prizes: approximately 93.8% of the prizes were awarded to male laureates (876 laureates), while only 6.2% were awarded to female laureates (58 laureates).

This disproportion reflects long-standing historical inequalities, which meant that women had limited access to education, research opportunities, and professional recognition in the 19th and early 20th centuries.

From an analytical perspective, the data not only reveals a performance gap but also highlights the systemic barriers and social norms that have limited women's participation in science and innovation. Even though the gap has gradually narrowed in recent decades, the overall distribution continues to be shaped **by this historical imbalance.**

This visualization makes it clear that the data on the Nobel Prizes is not only about scientific excellence, but also about who had the opportunity to make a contribution and gain recognition, making it an important reflection of broader social and cultural dynamics.

Breaking Barriers: Early Women Nobel Laureates

The history of the Nobel Prizes reflects broader societal trends, including the underrepresentation of women due to their limited access to education and career opportunities. For many decades, structural barriers prevented women worldwide from participating fully in scientific, literary, and peace-building efforts. Despite these challenges, a number of pioneering women managed to overcome these obstacles and make groundbreaking contributions worthy of a Nobel Prize.

Identifying the First Female Laureates

To better understand how women came to be recognized with Nobel Prizes, we are going to use simple filtering technique from pandas and attach it to the original DataFrame. Next, we are going to sort the data by year to receive the earliest entries.

This allows us to trace the origins of the recognition of women and examine the fields in which women first made their mark.

```
print("\nFirst five Women to recieve the Nobel Prize")
df_women = df_data[df_data.sex == 'Female']
df_women.sort_values('year', ascending=True)
print(df_women[columns_to_display].head(5))
```

The first five women to receive the Nobel Prize:

year	category	full_name	birth_date	birth_country	laureate_type	organization_name
1903	Physics	Marie Curie, née Sklodowska	1867-11-07	Russian Empire (Poland)	Individual	NaN
1905	Peace	Baroness Bertha Sophie Felicita...	1843-06-09	Austrian Empire (Czech Republic)	Individual	NaN
1909	Literature	Selma Ottilia Lovisa Lagerlöf	1858-11-20	Sweden	Individual	NaN
1911	Chemistry	Marie Curie, née Sklodowska	1867-11-07	Russian Empire (Poland)	Individual	Sorbonne University
1926	Literature	Grazia Deledda	1871-09-27	Italy	Individual	NaN

The results show that pioneering women such as Marie Curie stand out in particular and were even honored twice (in 1903 and 1911), underscoring their extraordinary contribution to science. The first female laureates were honored in various fields such as physics, peace research, literature, and chemistry, suggesting that women were making their mark in all disciplines despite social and economic barriers.

These early achievements are particularly significant because they have occurred at a time when women's participation in academia was extremely limited; as such, they represent not only scientific and cultural milestones but also symbolic breakthroughs for gender equality.

Multiple Nobel Prize Winners

While most Nobel laureates receive the award only once, a small but notable group has been honored multiple times for their achievements. One of the earliest examples is Marie Curie, who made history as the first woman to win a Nobel Prize and remains one of the few people to have been honored more than once. By looking at these multiple award winners, we can identify sustained virtue and long-term impact across different disciplines.

To detect the multiple winners, we are going to use the $duplicated()$ function to search for duplicate names in the $full\_name$ column within the dataset.

By filtering out these entries, we isolate all individuals and organizations that appear more than once, thereby identifying those who have been awarded the Nobel Prize multiple times.

```
# Multiple Winners

is_winner = df_data.duplicated(subset=['full_name'], keep=False)
# keep=False parameter prevents recurrence

multiple_winners = df_data[is_winner]
multiple_winners.sort_values('full_name')

print(multiple_winners[columns_to_display])
```

Laureates with More Than One Nobel Prize:

year	category	full_name	birth_date	birth_country	laureate_type	organization_name
1917	Peace	Comité international de la Croix Rouge	NaT	NaN	Organization	NaN
1944	Peace	Comité international de la Croix Rouge	NaT	NaN	Organization	NaN
1963	Peace	Comité international de la Croix Rouge	NaT	NaN	Organization	NaN
1958	Chemistry	Frederick Sanger	1918-08-13	United Kingdom	Individual	University of Cambridge
1980	Chemistry	Frederick Sanger	1918-08-13	United Kingdom	Individual	MRC Laboratory of Molecular Biology
1956	Physics	John Bardeen	1908-05-23	United States of America	Individual	University of Illinois
1972	Physics	John Bardeen	1908-05-23	United States of America	Individual	University of Illinois
1954	Chemistry	Linus Carl Pauling	1901-02-28	United States of America	Individual	California Institute of Technology (Caltech)
1962	Peace	Linus Carl Pauling	1901-02-28	United States of America	Individual	California Institute of Technology (Caltech)
1903	Physics	Marie Curie, née Sklodowska	1867-11-07	Russian Empire (Poland)	Individual	NaN
1911	Chemistry	Marie Curie, née Sklodowska	1867-11-07	Russian Empire (Poland)	Individual	Sorbonne University
1954	Peace	Office of the UN High Commissioner for Refugees	NaT	NaN	Organization	NaN
1981	Peace	Office of the UN High Commissioner for Refugees	NaT	NaN	Organization	NaN

The results show that 8 laureates have received the Nobel Prize more than once, including both individuals and organizations.

Notably, institutions such as International Committee of the Red Cross and United Nations High Commissioner for Refugees have been awarded multiple times, all within the Peace category, reflecting their ongoing humanitarian impact across different historical periods.

Among individuals, four scientists stand out for winning the Nobel Prize twice: Marie Curie, Frederick Sanger, John Bardeen, and Linus Pauling. Their achievements span in fields such as physics, chemistry, and even peace research, demonstrating exceptional versatility and a lasting scientific influence.

These scientists are not high achievers in their fields, but also rare examples of continuous innovation and global impact. They demonstrate that groundbreaking contributions can span multiple discoveries, disciplines, or decades.

One particularly notable figure among the multiple Nobel laureates is **Marie Curie.**

She received the Nobel Prize in Physics in 1903 for her groundbreaking research on radiation and later in 1911, in Chemistry, for her discovery of radium and polonium, making her still the **only person** to have won Nobel Prizes in two different scientific disciplines.

Despite her extraordinary achievements, Curie faced significant gender-based discrimination throughout her career, particularly in France, where women were often excluded from academic recognition and positions in institutions.

Nevertheless, she overcame these challenges with determination and left behind a legacy that not only revolutionized modern science but also continues to inspire women and people around the world who strive to overcome barriers and inequality.

Analysis of Nobel Prize Categories

The Nobel Prizes are awarded in various categories covering different fields of human achievement, ranging from scientific discoveries to humanitarian efforts. Fields such as physics, chemistry, and peace research—which have already been mentioned several times in the preceding sections—illustrate how the Nobel Prize recognizes both technical innovations and societal impacts.

In this section, we are going to explore how the prizes are distributed across these categories to better understand which fields have been honored most frequently throughout the history of the Nobel Prize.

Forming the DataFrame:

To examine this distribution, we group the dataset by the $Category$ field and count the number of prizes awarded in each category.

This aggregation step allows us to convert detailed data into a summarized format suitable for comparisons and visualizations.

```
df_category = df_data.groupby(["category"], as_index=False)
                     .agg({'prize': pd.Series.count})

df_category.sort_values('prize', ascending=False, inplace=True)
```

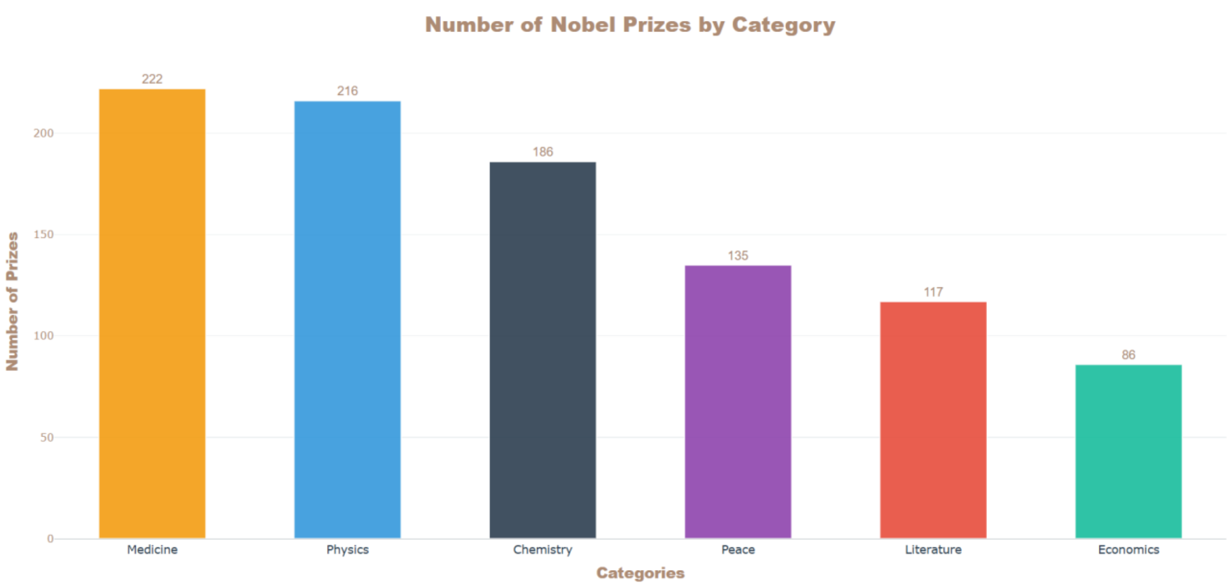
Creating bar Graph:

code:

```
category_colors = ['#2C3E50', '#18BC9C', '#E74C3C', '#F39C12', '#8E44AD',
'#3498DB']

fig = go.Figure(data=[go.Bar(
    x=df_category['category'],
    y=df_category['prize'],
    text=df_category['prize'],
    textposition='outside',
    textfont=dict(size=15, color='#AD8B73', family="Arial, sans-serif"),
    marker=dict(color=category_colors, opacity=0.9, line=dict(color='white',
width=1.5)),
    width=0.55
)])

fig.update_layout(
    title=dict(text="Number of Nobel Prizes by Category",
        font=dict(size=25, family="Arial Black", color="#AD8B73"),
        x=0.5, xanchor='center'),
```

```
    xaxis=dict(
        title=dict(text="Categories", font=dict(family="Arial Black", size=18,
color='#AD8B73')),
        tickfont=dict(size=14, color='#2C3E50'),
        categoryorder='total descending',
        showgrid=False,
        linecolor='#BDC3C7',
        linewidth=1
    ),

    yaxis=dict(
        title=dict(text="Number of Prizes",
        font=dict(family="Arial Black", size=18, color='#AD8B73')),
        tickfont=dict(size=13, color='#AD8B73'),
        showgrid=True, gridcolor='#ECF0F1', gridwidth=1,
        zeroline=True,
        zerolinecolor='#BDC3C7',
        zerolinewidth=1,
    ),

    plot_bgcolor='white',
    paper_bgcolor='white',
    margin=dict(t=80, b=40, l=60, r=40),
    showlegend=False
)

fig.show()
```

Graph:



The distribution shows that medicine and physics have received the most Nobel Prizes, closely followed by chemistry, highlighting the historical significance of scientific research and innovation.

In contrast, fewer prizes are awarded in categories such as peace, literature, and especially economics, which could be attributed both to their later introduction and to the more selective or subjective nature of these fields.

From an analytical perspective, this pattern suggests that the Nobel Prizes tend to focus more on scientific disciplines where discoveries are often cumulative and measurable, whereas contributions in the social sciences or humanities, while less frequent, are nonetheless of great significance.

Analysis of Gender Distribution Across Nobel Prize Categories

In addition to the overall gender distribution, it is important to examine how representation varies across the individual Nobel Prize categories. Based on this analysis, we can examine whether gender-based inequalities exist across all disciplines or are more pronounced or more improved in certain fields.

Forming the DataFrame:

To prepare the data for this analysis, we group the dataset by both $category$ and sex properties and count the number of awards given within each combination.

This approach uses grouping and aggregation techniques in pandas to create a structured dataset suitable for comparative and stacked visualizations.

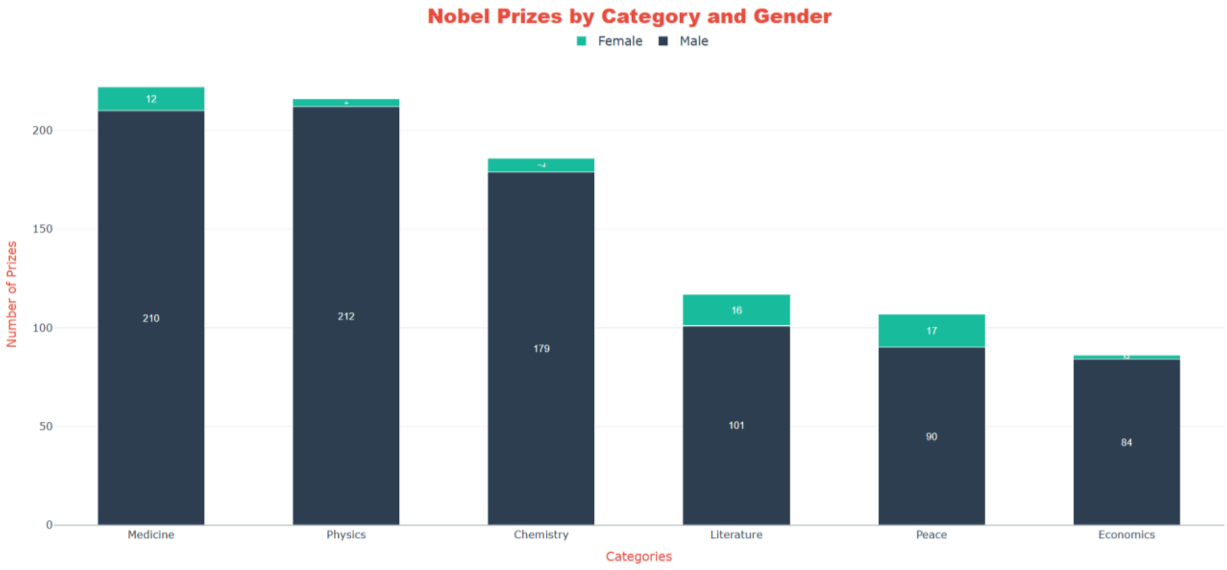
```
category_by_gender = df_data.groupby(['category', 'sex'], as_index=False)
                            .agg({'prize': pd.Series.count})

category_by_gender.sort_values('prize', ascending=False, inplace=True)
```

Creating the Stacked Bar Graph:

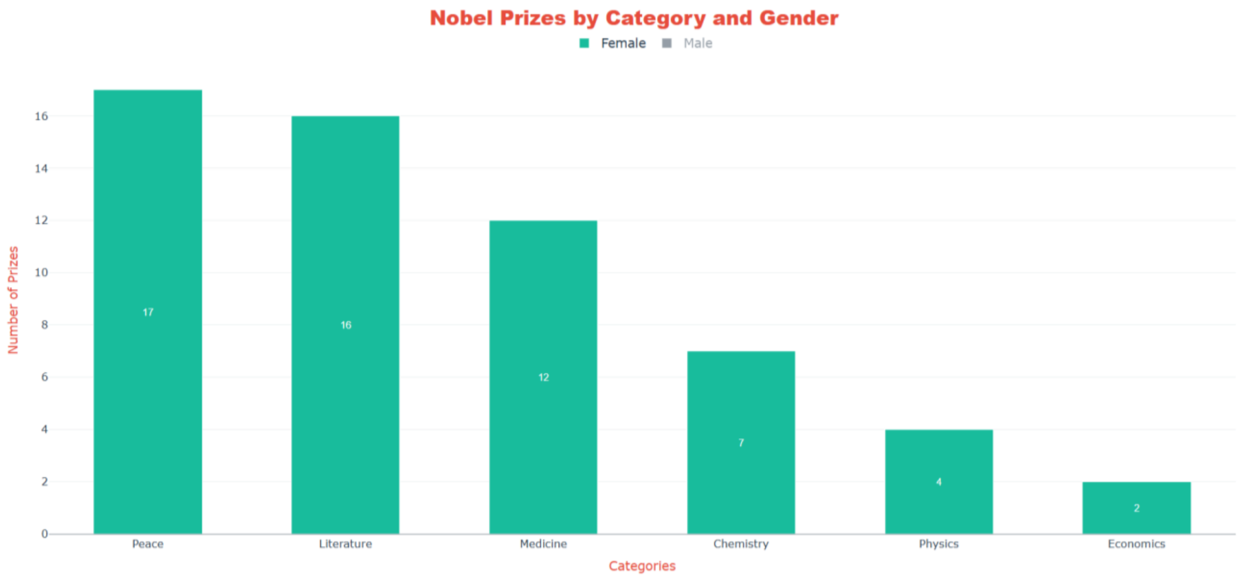
Code:

```python
colors = ['#2C3E50', '#18BC9C']  # Navy and Teal

fig = go.Figure()

# Iterating through the values for genders to generate the stacked sections
for i, gender in enumerate(category_by_gender['sex'].unique()):
    df_subset = category_by_gender[category_by_gender['sex'] == gender]
    fig.add_trace(go.Bar(
        x=df_subset['category'],
        y=df_subset['prize'],
        name=str(gender),  # naming the items
        marker=dict(color=colors[i % len(colors)],
                    line=dict(color='white', width=1)),  # Color and border styling
        text=df_subset['prize'],
        textposition='inside',
        insidetextanchor='middle',
        textfont=dict(size=12, color='white', family="Arial, sans-serif"),
        width=0.55  # a subtle width for the bar
    ))

fig.update_layout(
    barmode='stack', # so that we will have two parameters at bars
    title=dict(text="Nobel Prizes by Category and Gender",
               font=dict(size=24, family="Arial Black", color="#E74C3C"),
               pad=dict(b=40), x=0.5, xanchor='center'),

    legend=dict(orientation="h", yanchor="bottom", y=1.02,
                xanchor="center", x=0.5, font=dict(size=15, color='#2C3E50')),

    xaxis=dict(
        title=dict(text="Categories", font=dict(size=15, color='#E74C3C')),
        tickfont=dict(size=13, color='#2C3E50'),
        categoryorder='total descending',
        showgrid=False, linecolor='#BDC3C7', linewidth=1
    ),

    yaxis=dict(
        title=dict(text="Number of Prizes", font=dict(size=15, color='#E74C3C')),
        tickfont=dict(size=13, color='#2C3E50'),
        showgrid=True, gridcolor='#ECF0F1', gridwidth=1,
        zeroline=True, zerolinecolor='#BDC3C7', zerolinewidth=1,
    ),

    plot_bgcolor='white',
    paper_bgcolor='white',
    margin=dict(t=100, b=40, l=60, r=40)
)

fig.show()
```

Graph:



Filtering the Graph for Only Women:



Interactive graph Link: [Nobel Prizes by Category and Gender]

The stacked distribution highlights a consistent ongoing gender imbalance across all Nobel Prize categories, with male laureates dominating every field, particularly in physics and economics, where the proportion of women remains extremely low. However, fields such as peace studies and literature are characterized by a relatively higher proportion of women, suggesting that these fields have historically been more accessible or inclusive compared to highly technical scientific disciplines.

Notably, there is a comparatively high proportion of women in medicine field (12 female laureates), which may reflect the gradual increase in the number of women entering the fields of medicine and the life sciences over the course of the 20th century, as educational barriers gradually broke down. In contrast, the very low numbers in physics and economics point to long-standing structural and cultural barriers, including the underrepresentation of women in these fields and a slower pace of institutional change.

Overall, the data suggests that while progress has been made, gender equality in the awarding of the Nobel Prizes is still a work in progress, and future trends may show further improvement as more and more women enter traditionally male-dominated fields and assume leadership roles there.

Analysis of the Number of Nobel Prize Distribution Over Time

An analysis of the temporal distribution of Nobel Prizes provides valuable insights into historical trends, global events, and the evolution of scientific and social progress. By analyzing the number of prizes awarded annually between 1901 and 2020, we can identify periods of stability, upheaval, and growth in the awarding of the Nobel Prizes.

This historical perspective helps us understand how the Nobel Prize has evolved over the years and how external factors may have influenced the award patterns.

Forming the DataFrame:

To conduct this analysis, we aggregate the dataset by year and calculate the total number of awards granted annually. In addition, we apply a moving average (5-year window) to smooth out short-term fluctuations and highlight long-term trends in the data.

```
prize_per_year = df_data.groupby(by='year').count().prize
moving_average = prize_per_year.rolling(window=5).mean()
```

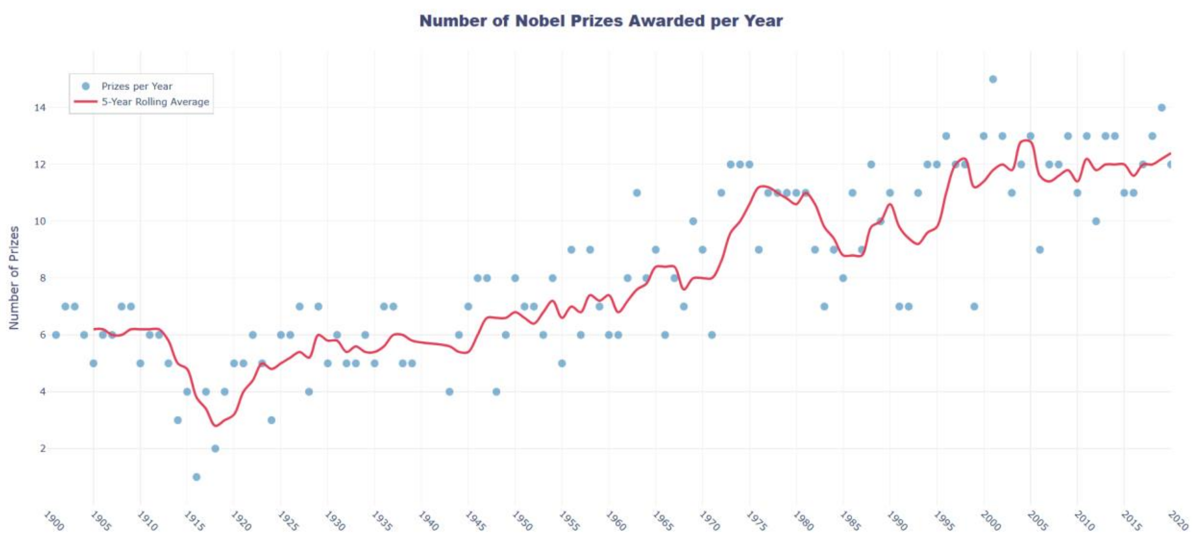
Creating scatter-line graph:

code:

```
# To create 5-year tick marks on the x-axis, we generate an array using NumPy:
np.arange(1900, 2021, step=5)

fig = go.Figure()

# Scatter graph (dots)
fig.add_trace(go.Scatter(
    x=prize_per_year.index,
    y=prize_per_year.values,
    mode='markers',
    name='Prizes per Year',
    marker=dict(color='#3282B8', size=11, opacity=0.6, line=dict(width=1,
color='white')),
    hovertemplate='<b>Year:</b> %{x}<br><b>Prizes:</b> %{y}<extra></extra>'
))

# Rolling Average line
fig.add_trace(go.Scatter(
    x=moving_average.index,
    y=moving_average.values,
    mode='lines',
    name='5-Year Rolling Average',
    line=dict(color='#E23E57', width=3, shape='spline'), # smoothes the line
    hoverinfo='skip'    # data points will focus when it hovers over them.
))

# Layout styling
fig.update_layout(
    title=dict(
        text='Number of Nobel Prizes Awarded per Year',
        font=dict(size=20, family="Arial Black", color="#424874"),
        y=0.95, x=0.5,
        xanchor='center', yanchor='top'
    ),
    xaxis=dict(title=dict(text="Year", font=dict(size=14, color='#424874'))),
    yaxis=dict(title=dict(text="Number of Prizes",
                          font=dict(size=14, color='#424874'))),
    template='plotly_white', # a modern white background
    hovermode='x unified',
    legend=dict(    # Legend allignments
        xanchor="left", yanchor="top",
        x=0.02, y=0.95,
        bgcolor="rgba(255, 255, 255, 0.8)",
        bordercolor="lightgrey",
        borderwidth=1
```

```
    ),
    margin=dict(l=60, r=40, t=80, b=60)
)

# Axes arrangements
fig.update_xaxes(
    tickangle=45,
    tickmode='linear',
    tick0=1900, dtick=5,
    showgrid=True, gridcolor='#f0f0f0',
    range=[1900, 2020]
)
fig.update_yaxes(showgrid=True, gridcolor='#f0f0f0')

# fig.write_html("interactive_graphs/nobel_prizes/prizes_per_year.html")

fig.show()
```

Graph:



Interactive Graph Link: [Number of Nobel Prizes Awarded per Year]()

The graph shows a clear long-term upward trend in the number of Nobel Prizes awarded each year, with the 5-year moving average rising steadily, particularly since the mid-20th century. A particularly striking decline can be observed between 1914 and 1918, when the number of awards dropped dramatically—this coincides with World War I, and similar collapses can also be seen during World War II, confirming that global conflicts had a direct impact on the awarding of the Nobel Prizes.

From an analytical perspective, the gradual increase in recent years can be attributed not only to scientific progress but also to changes in the award structure, as prizes are now more frequently shared among multiple laureates compared to previous decades. In addition, the introduction of new academic disciplines such as economics and the growing importance of collaborative, interdisciplinary research may have been contributed to the higher numbers observed in recent years.

Overall, the data suggests that trends in the allocation of the Nobel Prize are influenced by a combination of historical events, evolving scientific practices and institutional changes that reflect how global conditions and academic culture shape recognition over time.

Analysis of Nobel Prize Sharing Patterns Over Time

In this section, we are going to go beyond simply counting the number of Nobel Prizes awarded and focus on how these prizes have been distributed. By examining the distribution of prize stakes among laureates over time, we aim to determine whether collaboration and shared recognition have gained in importance.

Previously, we had observed a steady increase in the number of Nobel Prizes awarded each year—particularly since the mid-20th century. One possible explanation for this is that prizes are increasingly being shared among multiple contributors, reflecting the growing complexity and collaborative nature of modern science.

Forming the Dataset:

For this analysis, we group the dataset by year and calculate the average share of the prize money per winner. We then apply a moving window method to calculate a smoothed 5-year moving average, which allows us to identify long-term trends more clearly.

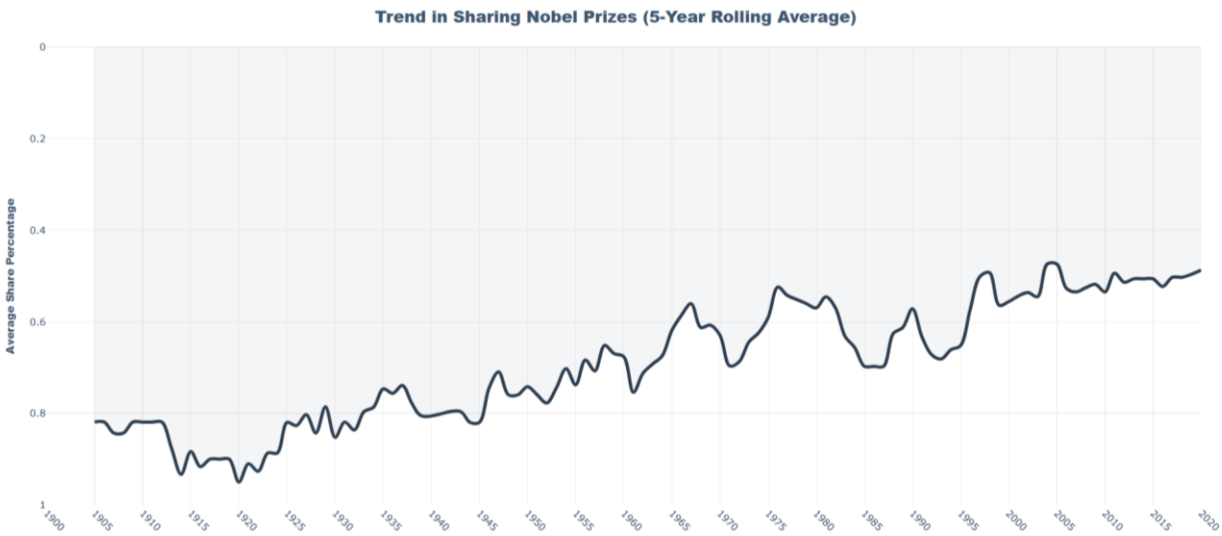
```
yearly_avg_share = df_data.groupby(by='year')
                          .agg({'share_pct': pd.Series.mean})

share_moving_average = yearly_avg_share.rolling(window=5).mean()
```

Creating Line Graph:

Code:

```
# Plotting the line with rolling average
fig.add_trace(go.Scatter(
    x=share_moving_average.index,
    y=share_moving_average['share_pct'],
    mode='lines',
    name='5-Year Moving Average',
    line=dict(color='#2c3e50', width=4, shape='spline'),
    fill='tozeroy',
    fillcolor='rgba(44, 62, 80, 0.05)'
))

fig.update_layout(
    title=dict(
        text='Trend in Sharing Nobel Prizes (5-Year Rolling Average)',
        font=dict(size=20, family="Arial Black", color="#3D566F"),
        x=0.5, y=0.95, xanchor='center', yanchor='top'),

    yaxis=dict(title=dict(
        text='Average Share Percentage',
        font=dict(size=13, family="Arial Black", color="#3D566F")),
        autorange="reversed", showgrid=True,gridcolor='#f0f0f0'),

    xaxis=dict(
        tickmode='linear', tick0=1900, dtick=5, tickangle=45,
        showgrid=True, gridcolor='#f0f0f0', range=[1900, 2020]),


    xaxis_title='',  # No need to name the x-axis.
    template='plotly_white',
    margin=dict(l=60, r=40, t=80, b=60),
    hovermode='x unified'
)

fig.show()
```

Graph:



Interactive graph link: Trend in Sharing Nobel Prizes

The graph shows a clear long-term shift from individual recognition toward collaboration, with the average share of the prize per laureate falling from about 0.8–0.9 at the beginning of the 20th century to about 0.5–0.6 in recent decades, suggesting that prizes are increasingly being shared among multiple laureates.

This trend has gone hand in hand with the rise of large-scale, interdisciplinary research subjects such as scientific breakthroughs, increasingly required teams, state-of-the-art laboratories, and international collaboration (for example, experiments in modern physics often involve hundreds of participants).

Interestingly, the fact that the figure has stabilized at around 0.5 in recent years could point to a practical upper limit dictated by the Nobel Prize rules (a maximum of three laureates per prize), meaning that while collaboration continues to increase, formal recognition remains limited.

Analysis of Nobel Prize Distribution by Country

In this section, we are going to explore how the Nobel Prizes are distributed among individual countries by identifying the 20 countries with the most laureates.

This analysis will help us understand which countries have had the greatest influence on the promotion of intellectual and humanitarian contribution in the past.

By examining geographical patterns, we can also gain insights into how factors such as education systems, research funding, and historical developments have shaped global scientific leadership.

Forming the DataFrame:

For this analysis, we group the dataset by the country of birth of each award winner and determine the total number of awards associated with each country.

The results are then sorted in descending order so that we can focus on the 20 countries with the most Nobel Prizes.

```
top20_countries = df_data.groupby(by='birth_country_current', as_index=False)
                         .agg({'prize': pd.Series.count})

top20_countries = top20_countries.sort_values('prize', ascending=False)[:20]
```

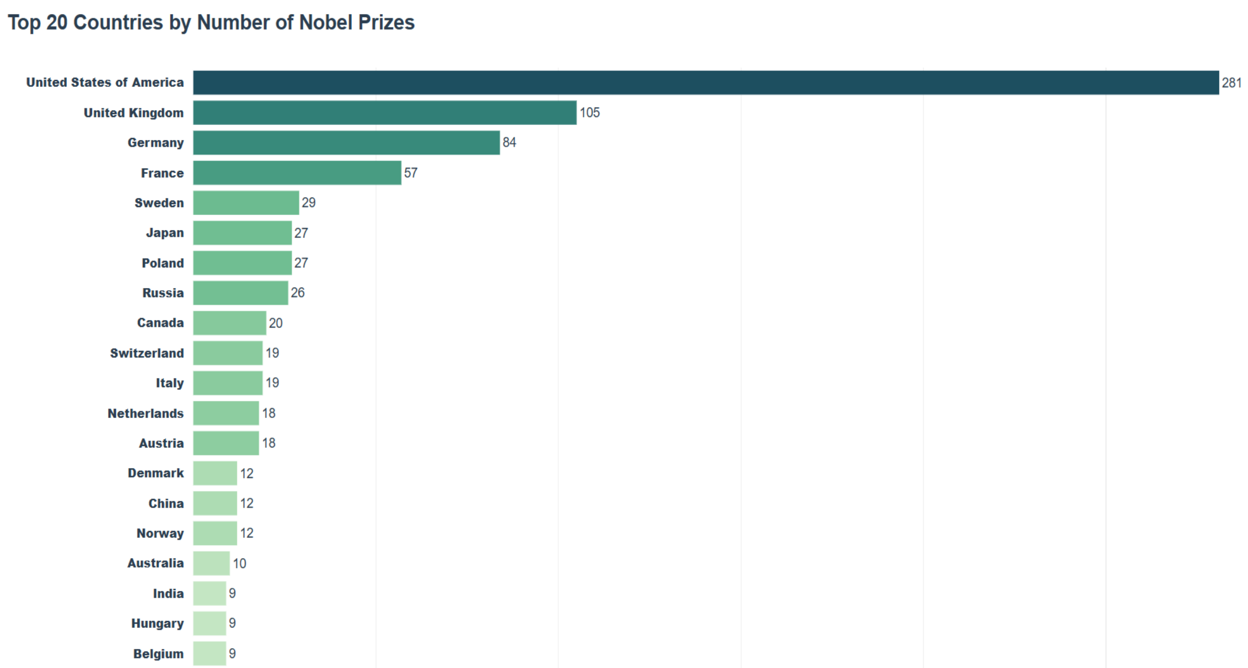
Creating the Bar Graph:

code:

```
# for the coloring of the bars we are going to use a logorithmic scale
top20_countries['log_prize'] = np.log10(top20_countries['prize'])

h_bar = px.bar(
    top20_countries,
    x='prize',
    y='birth_country_current',
    orientation='h',
    text='prize', color='log_prize',
    color_continuous_scale='blugrn',
    title='<b>Top 20 Countries by Number of Nobel Prizes</b>',
)

# Designing the Layout
h_bar.update_layout(
    plot_bgcolor='white', paper_bgcolor='white',
    font_family='Arial, Helvetica, sans-serif',
    title_x=0.01, title_font_size=22, font_color='#2c3e50',
```

```
    xaxis_title='', yaxis_title='', margin=dict(t=80, l=120, r=50, b=40),
    yaxis=dict(
        categoryorder='total ascending', showgrid=False,showline=False,
        ticks='outside', ticklen=7, tickcolor='white',
        tickfont=dict(family='Arial Black', size=13, color='#2c3e50')
    ),
    xaxis=dict(showgrid=True,showline=False, showticklabels=False),
    coloraxis_showscale=False, height=750
)

h_bar.update_traces( # Alignments for the labels and bars
    textposition='outside', textfont=dict(size=14, color='#2c3e50'),
    marker_line_width=0, cliponaxis=False,
    hovertemplate=(
        '<b style="font-size: 14px;">%{y}</b><br>' +
        '<span style="font-size: 13px;">Total Nobel Prizes: </span>' +
        '<b style="font-size: 14px;">%{x}</b>' +
        '<extra></extra>' ),
)

h_bar.write_html("interactive_graphs/nobel_prizes/top_20_countries.html")
```

Graph:



Graph Link: Top 20 Countries by Number of Nobel Prizes

The graph shows a striking concentration of Nobel Prizes, with the United States leading by a wide margin (281 prizes) and far ahead of the United Kingdom (105) and Germany (84), illustrating a significant imbalance in global scientific achievement.

This dominant position can be explained in part by the post-World War II scientific migration, during which many European scientists moved to the United States, as well as by sustained investment in research and development—which today accounts for nearly 30% of global Research and Development (R&D) spending.

Interestingly, smaller European countries such as Switzerland and Sweden still rank among the top performers despite their size, suggesting that strong academic institutions and innovation-oriented policies can outweigh population size when it comes to making high-impact contributions.

Analysis of Global Nobel Prize Distribution Using Choropleth Mapping

In this section, we are going to expand our analysis to a global level by illustrating the geographical distribution of Nobel laureates worldwide. Plotly's choropleth map provides an intuitive way to see how Nobel Prizes are distributed among all countries and to identify regional patterns and differences.

This approach allows us to go beyond simple rankings and examine the spatial dimension of the Nobel Prizes, revealing how different parts of the world contribute to global scientific and societal impact.

Forming the DataFrame:

To create this visualization, we group the dataset by both country name and ISO code, since Plotly's choropleth maps rely on ISO codes to correctly assign the data to the countries.

We then count the total number of awards per country and apply a logarithmic transformation to the values to even out the color distribution and prevent particularly successful countries from overshadowing others.

```
df_countries = df_data.groupby(['birth_country_current', 'ISO'],
as_index=False)                .agg({'prize': pd.Series.count})
```

```
df_countries.sort_values('prize', ascending=False)

# Add logarithmic column to evenly distribute the color gradient
df_countries['log_prize'] = np.log10(df_countries['prize'])
```

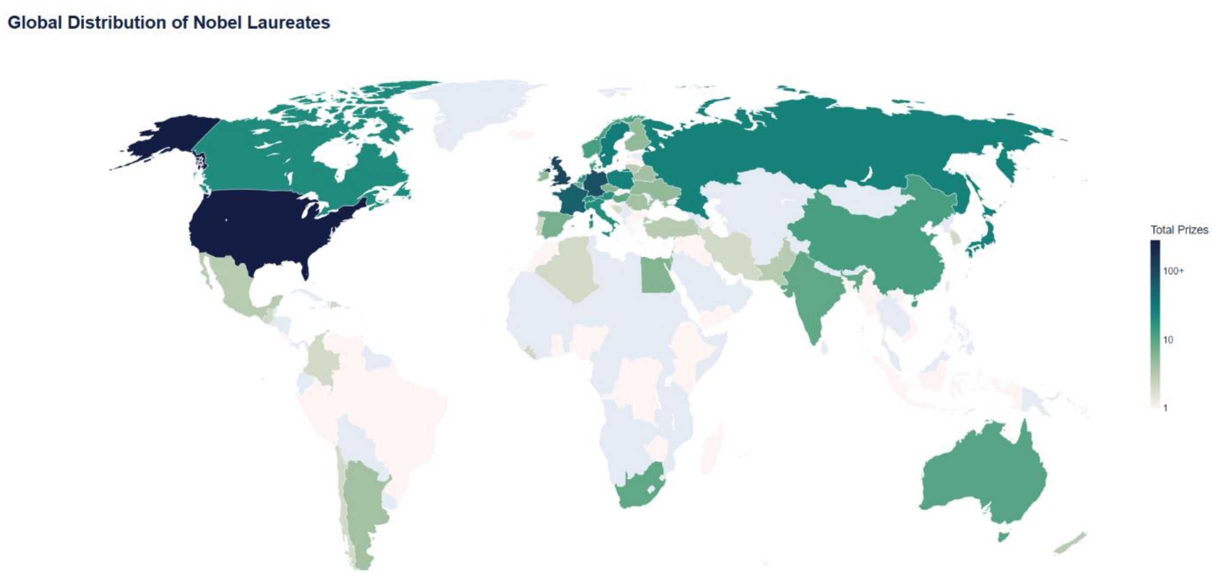
Creating World Map Graph:

code:

```
# Plotting the Choropleth
world_map = px.choropleth(
    df_countries,
    locations='ISO',
    color='log_prize',
    hover_name='birth_country_current',
    custom_data=['prize'],
    color_continuous_scale='tempo',
    title='<b style="color:#152346;">Global Distribution of Nobel Laureates</b>'
)

# Layout adjustments
world_map.update_layout(
    font_family='Arial, Helvetica, sans-serif',
    font_color='#2c3e50',
    title_font_size=22,
    title_x=0.01,
    margin=dict(t=80, l=0, r=0, b=0),
    geo=dict(
        showframe=False,  # remove the border around the world
        showcoastlines=False,  # remove the thick lines
        projection_type='natural earth',  # an earth projection
        bgcolor='rgba(0,0,0,0)',  # transparent background
        lakecolor='#ffffff',  # fill lakes with white to blend in cleanly
    ),

    coloraxis_colorbar=dict(
        title=dict(text="Total Prizes", font=dict(size=14, color='#2c3e50')),
        thicknessmode="pixels", thickness=12,
        lenmode="pixels", len=250,
        yanchor="middle", y=0.5,
        xanchor="left", x=0.95,
        tickvals=[0, 1, 2],
        ticktext=['1', '10', '100+'],
        tickfont=dict(size=12, color='#2c3e50')
    )
)

world_map.update_traces(
```

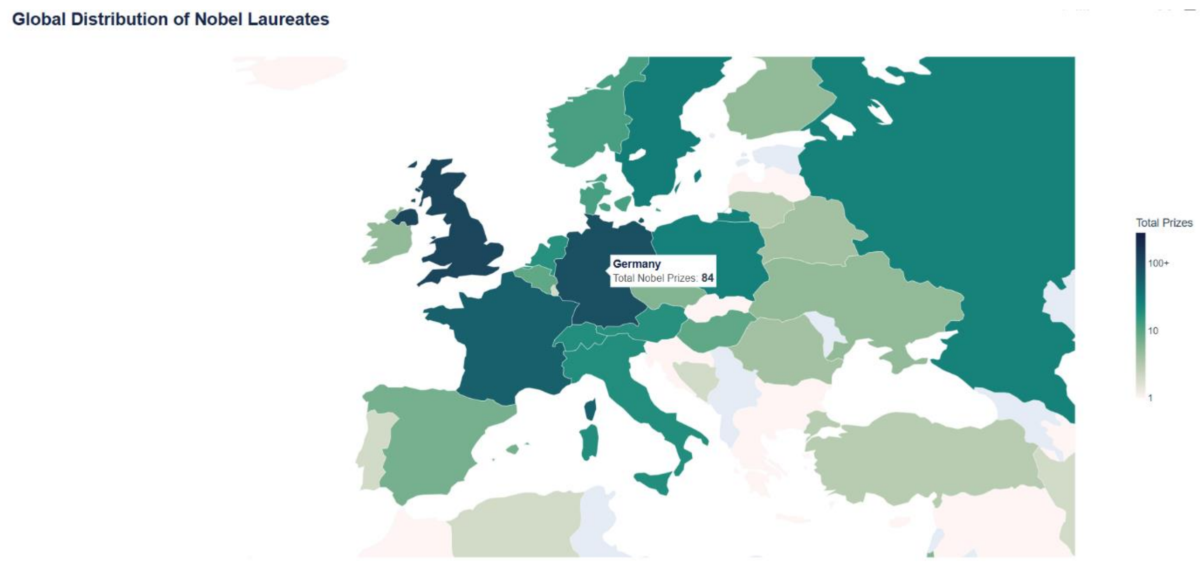
```
    marker_line_color='#ffffff',  # a subtle white borders between countries
    marker_line_width=0.4,  # but hey must be thin

    # Hover arrangements:
    hovertemplate=(
            '<b style="font-size: 14px; color: #152346;">%{hovertext}</b><br>' +
            '<span style="font-size: 13px; color: #555;">Total Nobel Prizes:
</span>' +
            '<b style="font-size: 14px; color: #2c3e50;">%{customdata[0]}</b>' +
            '<extra></extra>'
    ),

    hoverlabel=dict(bgcolor="white", font_size=14, font_family="Arial",
bordercolor="#e2e8f0")
)

world_map.show()
```

Graph:



Interactive Map Link: [Global Distribution of Nobel Laureates]()

Europe:



The choropleth map shows an extremely uneven global distribution of Nobel Prizes, with a strong concentration in North America and Western Europe, while large regions such as Africa and parts of Asia are poorly represented.

The United States stands out significantly with 281 awards, placing it far ahead of other countries; this reflects not only its research capabilities but also its historical role in attracting global scientific talent.

Interestingly, countries such as China and India, despite their large populations, have relatively modest numbers (about 12 and 9, respectively), suggesting that population size alone does not directly lead to Nobel Prize-level achievements and that long-term investment in research ecosystems is a decisive factor.

Overall, the map highlights a structural imbalance in global knowledge production, which is likely attributable to differences in education systems, funding, and international cooperation networks, although this pattern may gradually shift as emerging economies continue to expand their scientific infrastructure.

Analysis of Nobel Prize Category Distribution Across Top Countries

In this section, we take it a step further and examine not only which countries lead in Nobel Prizes, but also how these prizes are distributed across the various categories within the top 20 countries. This allows us to determine whether certain countries have specialized in specific fields such as physics, medicine, or peace research.

To illustrate this, we use a stacked bar chart that breaks down the total number of award winners from each country by category, providing a more nuanced overview of national strengths and contributions.

Forming the DataFrame:

To perform this analysis, we first group the dataset by both country and category and count the number of awards in each combination to capture the results at the category level.

We then combine this result with the previously calculated dataset for the top 20 countries, so that we can retain both the category-specific figures and the total number of awards for proper sorting and comparison.

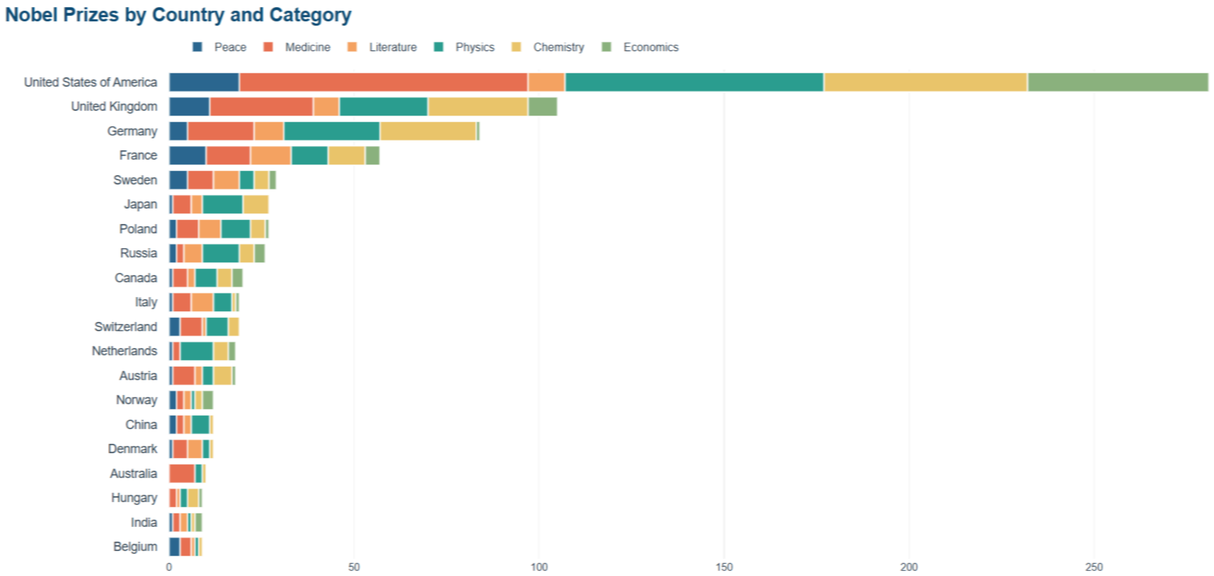
```
prize_by_category_and_country = df_data
                   .groupby(['birth_country_current', 'category'], as_index=False)
                   .agg({'prize': pd.Series.count})

# Next, we merge this DataFrame with the top20 countries we created previously.
# That way we get the total number of prizes in a single column too.
# This is important since we want to control the order for our bar chart.

top20_countries = df_data.groupby(by='birth_country_current', as_index=False)
                         .agg({'prize': pd.Series.count})

top20_countries.sort_values('prize', ascending=False)[:20]


merged_df = pd.merge(prize_by_category_and_country, top20_countries,
                     on='birth_country_current')


# changing the names of columns
merged_df.columns=  ['birth_country_current','category','cat_prize','total_prize']

merged_df.sort_values(by='total_prize', inplace=True)
```

Creating Stacked Bar Graph:

Code:

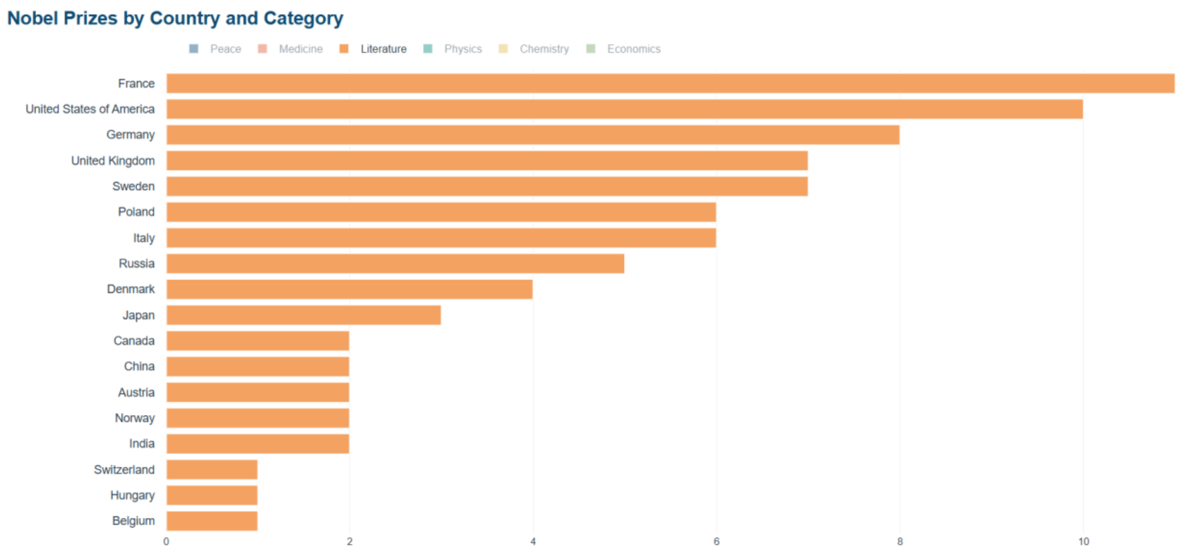
```
# color palette for the 6 categories
elegant_palette = ['#2A668F', '#E76F51', '#F4A261', '#2A9D8F', '#E9C46A',
'#8AB17D']

# Plotting the bar chart
cat_cntry_bar = px.bar(
    df_category_and_country,
    x='cat_prize',
    y='birth_country_current',
    color='category',
    orientation='h',
    color_discrete_sequence=elegant_palette,
    title='<b style="color:#0F4C75;" >Nobel Prizes by Country and Category</b>'
)

# Layout Design
cat_cntry_bar.update_layout(
    plot_bgcolor='white', paper_bgcolor='white',
    font_family='Arial, Helvetica, sans-serif', font_color='#2c3e50',
    title_font_size=22, title_x=0.01,
    margin=dict(t=110, l=140, r=50, b=40),  # allignments to fit the new legend

    yaxis=dict(
        categoryorder='total ascending',
        showgrid=False, showline=False,
        ticks='outside', ticklen=10, tickcolor='white',
        tickfont=dict(size=14, color='#2c3e50')),

    xaxis=dict(showgrid=True, gridcolor='#f0f0f0',showline=False, zeroline=False),
    legend=dict(title='', orientation='h', yanchor='bottom',xanchor='left',
        font=dict(size=13, color='#2c3e50'))
)

cat_cntry_bar.update_traces(
    marker_line_color='white',
    marker_line_width=1.5,
    hovertemplate=(
            '<b style="font-size: 14px color:#0F4C75;">%{y}</b><br>' +
            '<span style="font-size: 13px; color: #555;">%{data.name}: </span>' +
            '<b style="font-size: 14px; color: #2c3e50;">%{x}</b>'),
    hoverlabel=dict(bgcolor="white", font_family="Arial", bordercolor="#e2e8f0")
)

cat_cntry.show()
```

Graph:

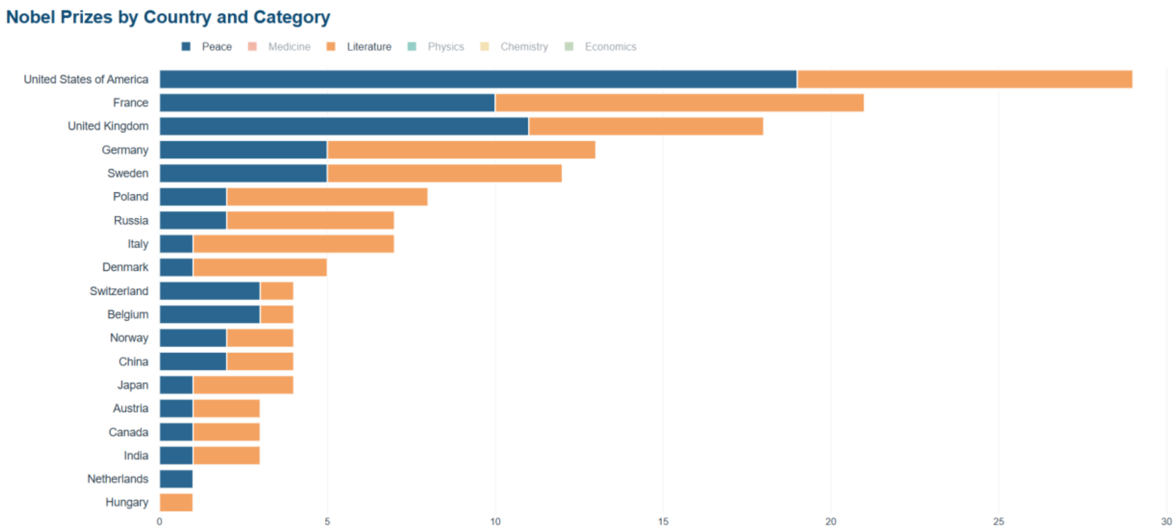


Interactive Graph Link: [Nobel Prizes by Country and Category]()

Filtering only for Literature:



Filtering for Peace and Literature:



The stacked bar chart shows that while the United States dominates overall, its leading position is not evenly distributed across all categories—in particular, it is not the leading country in the field of literature, where France clearly stands out due to its strong cultural and intellectual tradition.

Compared to the United States, Germany and Japan appear to be relatively weaker, particularly in the fields of medicine and peace, while, interestingly, Germany outperforms the United Kingdom in chemistry and France outperforms Germany in literature and peace, highlighting the different national specializations.

A closer look also reveals that countries like Australia place a strong emphasis on medicine, whereas the Netherlands is largely shaped by the Physics which accounts for a significant share of its total output.

Analysis of Cumulative Nobel Prize Patterns by Country Over Time

In this section, we are going to examine how global leadership in Nobel Prizes has evolved over time, taking into account that the influence of individual countries varies across different historical periods.

By tracking the evolution of the awards year by year, we can identify when significant changes took place—for example, the point at which the United States surpassed the traditionally dominant European nations.

This analysis provides answers to key questions about historical scientific leadership and it highlights which countries were leaders in earlier decades and how global dynamics have shifted over the course of the 20th century.

Creating the DataFrame:

To analyze trends over time, we first group the dataset by country and year and count the number of awards given to each country each year. This allows us to create a time series that tracks how many awards each country received annually.

Next, we sort the data chronologically and calculate the cumulative total of the prizes for each country. This step is crucial because it converts the annual figures into a running total, allowing us to observe how each country's total number of Nobel Prizes grows over time and when one country overtakes another.

```
prize_by_year = df_data
                    .groupby(by=['birth_country_current','year'], as_index=False)
                    .count()

prize_by_year = prize_by_year.sort_values('year')
prize_by_year = prize_by_year[['year', 'birth_country_current', 'prize']]
```

Then we create a series for the cumulative sums of the number of received prizes.

```
cumulative_prizes = prize_by_year
                                .groupby(by=['birth_country_current', 'year'])
                                .sum().groupby(level=[0])
                                .cumsum()

cumulative_prizes.reset_index(inplace=True)
```

The data is first aggregated using $.sum()$ to ensure that the price values are correctly summed and then regrouped by country $(level = [0])$ to display each country's time series separately.

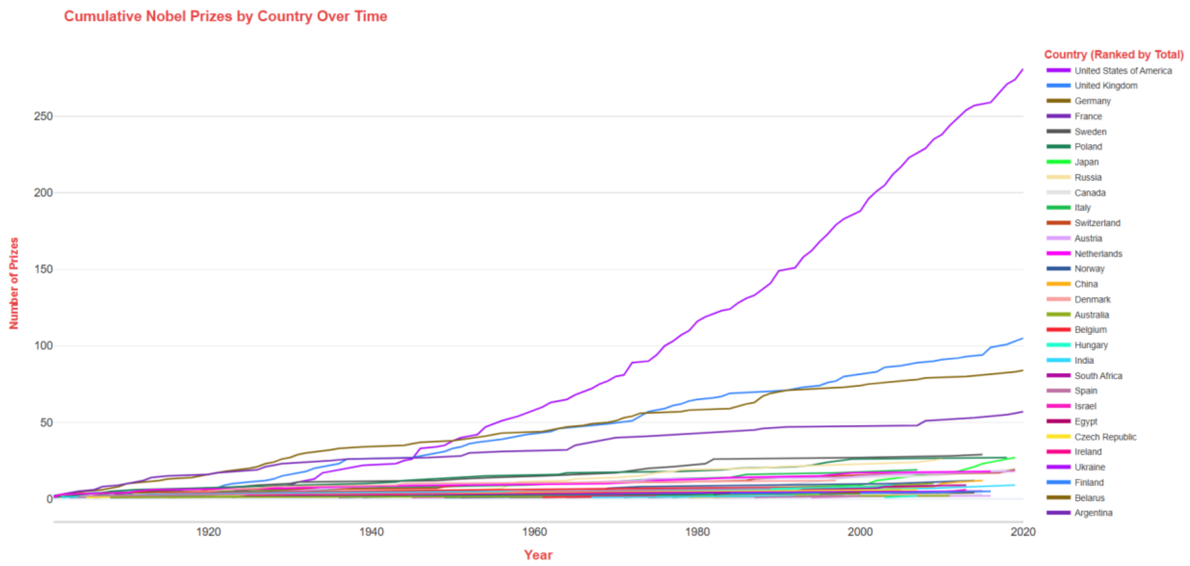
Finally, $.cumsum()$ is applied to calculate a cumulative sum of prices over time, making it easy to track growth and identify when countries overtake one another.

Creating the Collective Line Graph:

code:

```
We group by country, find their max prizes, sort descending, and get the list of
names.
country_totals = cumulative_prizes.groupby('birth_country_current')['prize'].max()
country_totals.sort_values(ascending=False)
ordered_countries = country_totals.index.tolist()

country_line = px.line(
    cumulative_prizes,
    x='year',
    y='prize',
    color='birth_country_current',
    hover_name='birth_country_current',
    hover_data={'birth_country_current': False},
    category_orders={"birth_country_current": ordered_countries},
    color_discrete_sequence=px.colors.qualitative.Alphabet
)

country_line.update_layout(
    title="<b>Cumulative Nobel Prizes by Country Over Time</b>",
    template='plotly_white',  # Removing the default  background and gridlines
    font=dict(family="Helvetica, Arial, sans-serif", size=14, color="#333333"),
    title_font=dict(size=18, color="#E84545"),
    xaxis_title='<b style="color:#E84545;">Year</b>',
    yaxis_title='<b style="font-size: 14px; color:#E84545;">Number of Prizes</b>',
    legend_title='<b style="font-size: 14px; color:#E84545;">Country (Ranked by
Total)</b>',

    legend=dict(
        yanchor="top", xanchor="left",
        y=1, x=1.02, # positioning the legend outside the plotting area
        font=dict(size=11),
        itemsizing='constant'),

    margin=dict(l=40, r=40, t=60, b=40)
)

country_line.update_xaxes(showgrid=False, showline=True, linewidth=1.5,
linecolor='lightgray')
country_line.update_yaxes(showgrid=True, gridwidth=1, gridcolor='#E5E5E5',
showline=False)

country_line.write_html("interactive_graphs/nobel_prizes/prize_by_country_over_yea
rs.html")

# country_line.show()
```

Graph:



Interactive Graph Link: [Cumulative Nobel Prizes by Country Over Time]

The cumulative trends clearly show a significant historical shift in global scientific leadership, in which the former dominance of European countries—particularly Germany, France, and the United Kingdom—has gradually given way to a strong and sustained rise of the United States, especially after the 1940s, which eventually surpassed all others by a wide margin (and reaching 280+ prizes).

This turning point largely coincided with World War II and its aftermath, when many leading European scientists emigrated to the United States and the country significantly increased its investment in research, universities, and innovation systems.

Meanwhile, countries such as Sweden and Poland are experiencing steady and consistent, growth, while nations like Japan and China are showing a late but accelerating upward trend, reasoned to their recent expansion of scientific infrastructure and increasing global integration.

Overall, the chart illustrates that leadership in Nobel Prizes is not static, but is shaped by historical events, investments in educational research and development, and countries' ability to attract and retain top talent over the long term.

Analysis of Prize Distribution Across Leading Organizations

In this section, we are going to shift our focus from countries to institutions and examine how the Nobel Prizes are distributed among organizations such as universities, research institutes, and laboratories.

From this perspective, we can identify the key institutions that have consistently contributed to revolutionary discoveries and global scientific progress.

By examining leading organizations, we can better understand where groundbreaking research is taking place and how certain institutions have established themselves as long-term centers of excellence.

Creating the DataFrame:

To conduct this analysis, we determine the number of Nobel Prizes associated with each organization, select the top 20, and then rank them to provide a clear and meaningful overview.

```
top20_orgs = df_data.organization_name.value_counts()[:20]
top20_orgs.sort_values(ascending=True, inplace=True)
```

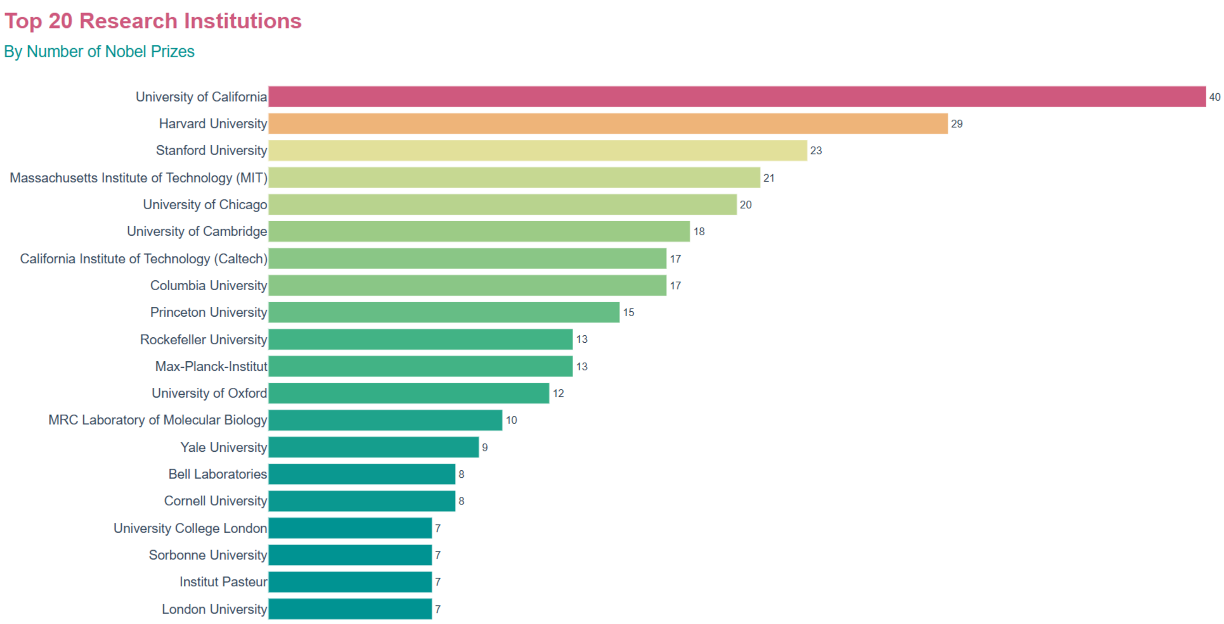
Creating Horizontal Bar Graph:

code:

```
# Create a modern, minimalist figure
org_bar = go.Figure()

org_bar.add_trace(go.Bar(
    x=top20_orgs.values,
    y=top20_orgs.index,
    orientation='h',
    text=top20_orgs.values,
    textposition='outside', # Puts the number neatly outside the bar
    textfont=dict(family="Arial, sans-serif", size=12, color='#2C3E50'),
    marker=dict(
        color=top20_orgs.values,
        colorscale='temps',
        line=dict(color='rgba(255, 255, 255, 1)', width=1)
    )
))
```

```
org_bar.update_layout(
    plot_bgcolor='white',
    paper_bgcolor='white',
    title=dict(
        text='<b>Top 20 Research Institutions</b><br><span style="font-size: 18px;
color: #009392;">By Number of Nobel Prizes</span>',
        x=0.02, y=0.95, # Align slightly to the left
        font=dict(family="Arial, sans-serif", size=25, color='#cf597e')),

    yaxis=dict(
        showgrid=False,showline=False, title='',
        tickfont=dict(family="Arial, sans-serif", size=15, color='#34495E')),

    xaxis=dict(showgrid=False, showline=False, showticklabels=False,
zeroline=False, title=''),
    margin=dict(l=200, r=50, t=100, b=30),
    height=750
)

org_bar.show()
```

Graph:



Interactive graph link: [Top 20 Research Institutions]()

The chart shows a strong concentration of Nobel Prizes at elite universities, with the University of California leading the way with 40 prizes, followed by Harvard and Stanford Universities with respectively 29 and 23 Nobel Prizes, illustrating the dominance of leading U.S. research universities.

This pattern reflects the enormous research funding and extensive infrastructure in the United States, where universities such as MIT and Stanford benefit from close ties to industry and the government—in fact, a significant portion of global R&D spending is concentrated in the U.S., which directly contributes to groundbreaking discoveries.

Interestingly, while American institutions dominate the top rankings, distinguished European centers such as Max-Planck, Cambridge and Oxford remain highly competitive, suggesting that long-standing academic traditions combined with sustained investment are key factors in achieving Nobel Prize-level success.

Analysis of Prize Distribution Across Leading Research Cities

In this section, we are going to focus on geographical research hubs and examine which cities have produced the most Nobel Prize-winning work. By examining the locations of the institutions with which the award winners are affiliated, it is possible to identify global hubs where innovation and academic excellence are most concentrated.

This perspective highlights how certain cities—which often host renowned universities and research laboratories—become hubs of intellectual activity, attract talent, and foster groundbreaking discoveries.

Creating the DataFrame

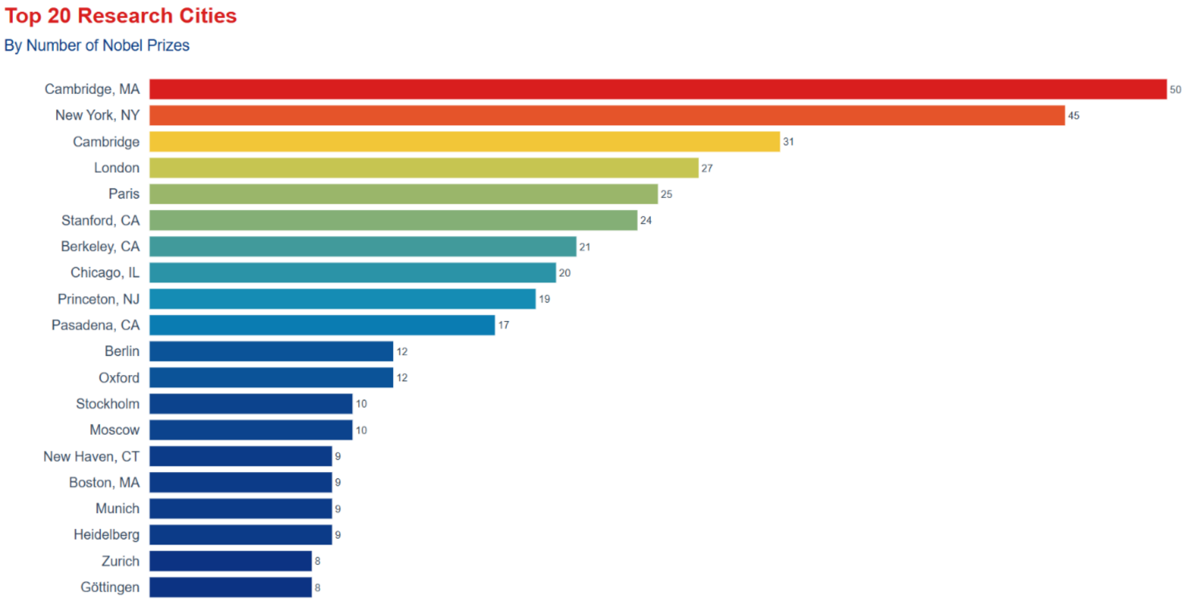
For this analysis, we determine the frequency of Nobel laureates' affiliations by city, select the 20 most frequent ones, and sort them to provide a clear overview.

```
top20_org_cities = df_data.organization_city.value_counts()[:20]
top20_org_cities.sort_values(ascending=True, inplace=True)
```

Creating the Bar Graph:

The same methodology used in the previous analysis is applied here, with minor adjustments to adjust the context of city-level data.

Graph:



Graph Link: [Top 20 Research Cities]

The graph illustrates a strong concentration of successful research in a few global hubs, with Cambridge, MA and New York leading by a wide margin, reflecting the high concentration of elite institutions such as Harvard and MIT in the Boston area.

This pattern illustrates the concept of innovation centers, in which the geographical proximity of top universities, funding agencies, and the business community creates an ecosystem that fosters impactful research. Regions such as Boston and Silicon Valley rank among the global leaders in terms of academic research.

On the other hand, European cities such as Cambridge, London, and Paris remain highly competitive, suggesting that traditional academic hubs continue to play an important role, even though overall dominance has gradually shifted to research centers in the United States. Interestingly, while Germany ranks among the leading countries overall, its research landscape is more dispersed: Berlin ranks only 11th, and other cities such as Munich and Heidelberg also appear on the list, suggesting a more decentralized and regionally balanced research ecosystem rather than a single dominant hub.

Analysis of Birth Cities of Nobel Laureates

In this section, we are going to turn our focus from the places where research is conducted to the birthplaces of Nobel laureates. By analyzing places of birth, we aim to uncover geographical patterns in the origins of outstanding talent and identify cities that have produced a particularly large number of Nobel laureates in the past.

This perspective offers insight into living conditions during early childhood, educational systems, and cultural factors that can contribute to the development of future high achievers.

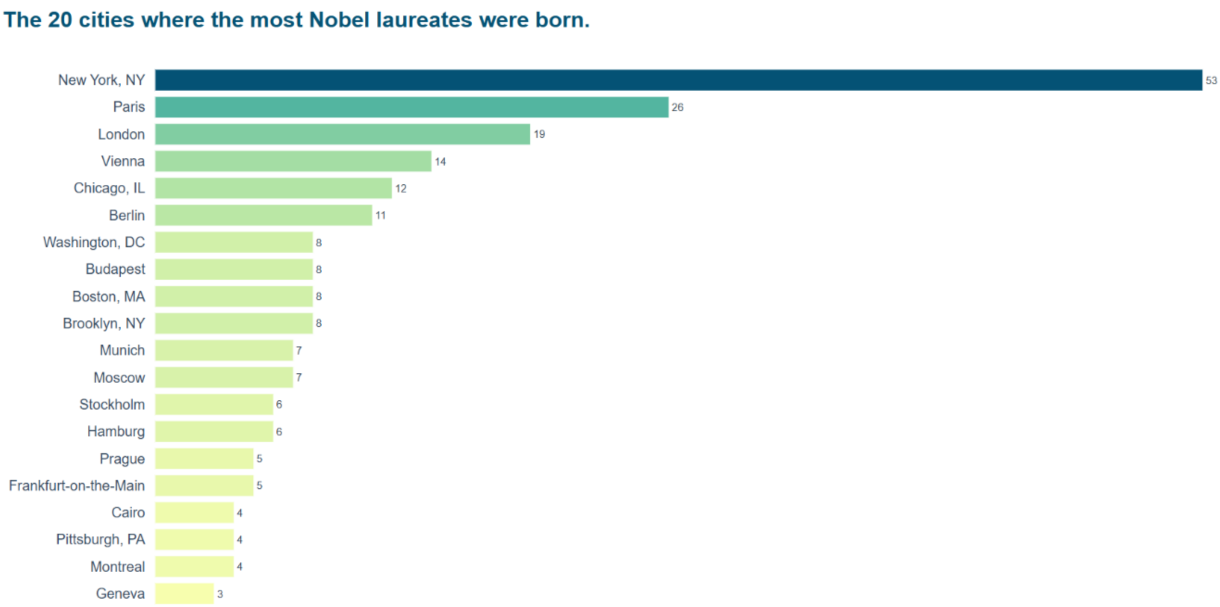
Preparing the DataFrame:

To conduct this analysis, we record the frequency of award winners by place of birth, select the 20 most common locations, and sort the results to provide a clear and structured overview.

```
top20_birth_cities = df_data.birth_city.value_counts()[:20]
top20_birth_cities.sort_values(ascending=True, inplace=True)
```

Creating the Bar Graph:

The same methodology used in the previous analysis is applied here, with minor adjustments to adjust the new analysis.



Graph Link: Top 20 Birth Cities

The graph shows that New York (53) stands out as the dominant birthplace of Nobel laureates, far outpacing cities such as Paris (26) and London (19), suggesting that large, diverse metropolitan areas may play a crucial role in fostering early intellectual development.

Interestingly, compared to earlier analyses, there is a clear distinction between the places where talent emerges and those where it flourishes, as many historically significant European cities (e.g., Vienna, Berlin, Budapest) play a prominent role here despite the subsequent dominance of U.S. research institutions—which underscores the influence of **migration and global academic mobility.**

This pattern reflects broader trends of the 20th century, when political events (such as World War II) and educational opportunities led many scientists born in Europe to move to research centers like the United States, reinforcing the notion that **talent is distributed globally, but opportunities are geographically concentrated.**

Analysis of All Countries and Organizations Using Hierarchical Visualization

In this section, we examine how Nobel Prize-winning research is distributed across countries, cities, and institutions. The hierarchical organization of the data allows us to better understand in which geographic regions significant scientific contributions are concentrated.

To achieve this, we are going to use a sunburst chart, which enables us to present multiple levels of information in a single, intuitive graphic.

Sunburst chart will enable us to break down each country into its contributing cities, and further into the organizations within those cities. This visualization type is particularly effective here, as it clearly illustrates how scientific achievements are layered geographically—from global to local scales.

Preparing the DataFrame:

Before creating the visualization, we aggregate the dataset by grouping it by country, city, and organization, and counting the number of awards associated with each combination.

In addition, we apply a logarithmic transformation to the number of prizes to improve the color scaling, thereby ensuring a more balanced and visually meaningful representation of both major and minor contributors.

```
country_city_org = (df_data
                    .groupby(by=['organization_country', 'organization_city',
'organization_name'], as_index=False)
                    .agg({'prize': pd.Series.count}))

country_city_org = country_city_org.sort_values('prize', ascending=False)

# Just for coloring, so that we have a balanced coloring function
country_city_org['log_prize'] = np.log10(country_city_org['prize'])
```

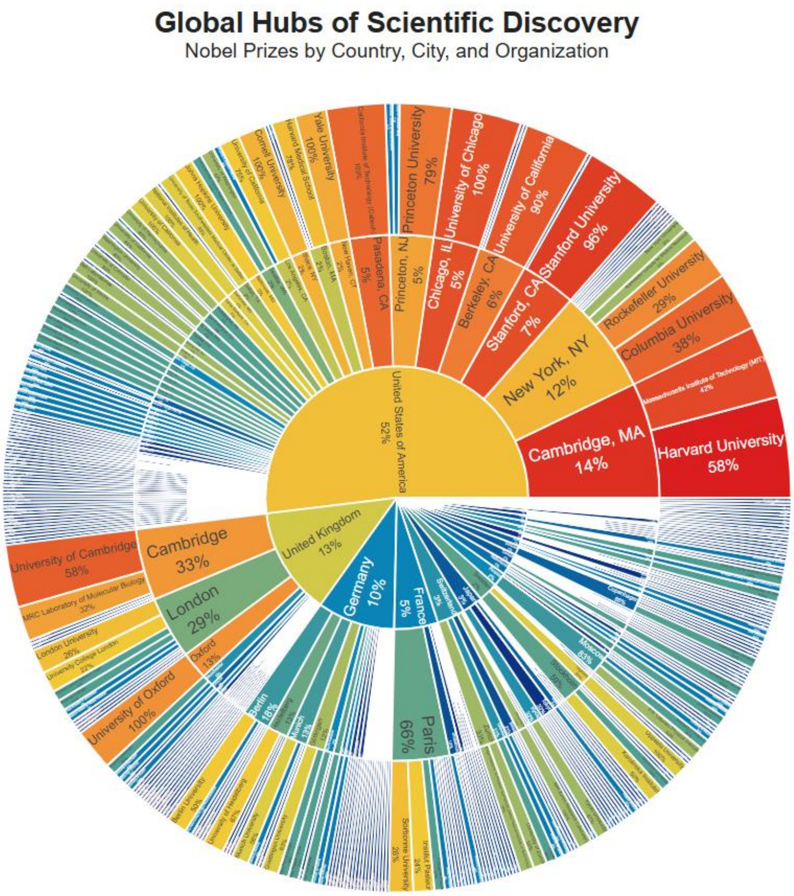
Creating Sunburst Chart:

code:

```
burst = px.sunburst(
    country_city_org,
    path=['organization_country', 'organization_city', 'organization_name'],
    values='prize',
    color='log_prize',
    color_continuous_scale='portland',
    title='<b>Global Hubs of Scientific Discovery</b><br><sup>Nobel Prizes by
Country, City, and Organization</sup>'
)

burst.update_layout(
    font=dict(family="Helvetica, Arial, sans-serif", size=13, color="#444444"),
    title_font=dict(size=22, color="#222222"),
    title_x=0.5, title_y=0.95,
    margin=dict(t=90, l=20, r=20, b=20),
    paper_bgcolor="white",
    coloraxis_showscale=False
)
```

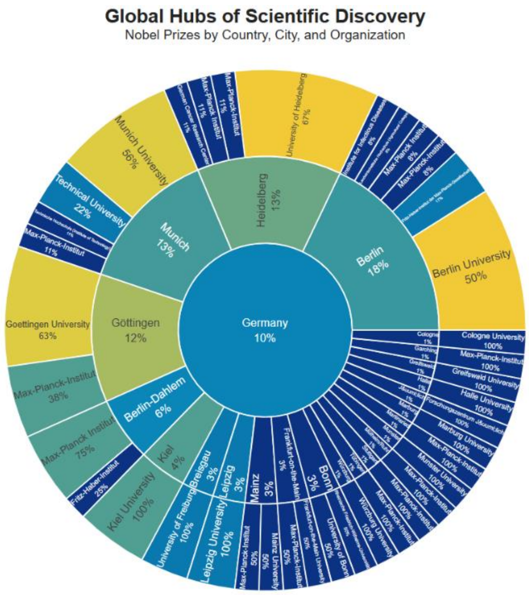
```
burst.update_traces(
    textinfo='label+percent parent', insidetextorientation='radial',
    marker=dict(line=dict(color='#FFFFFF', width=1.5)),
    hovertemplate=(
        '<b>%{label}</b><br>'
        'Total Discoveries: <b>%{value}</b><br>'
        '<i>%{percentParent:.1%} of parent category</i>'
        '<extra></extra>'
    )
)


burst.show()
```

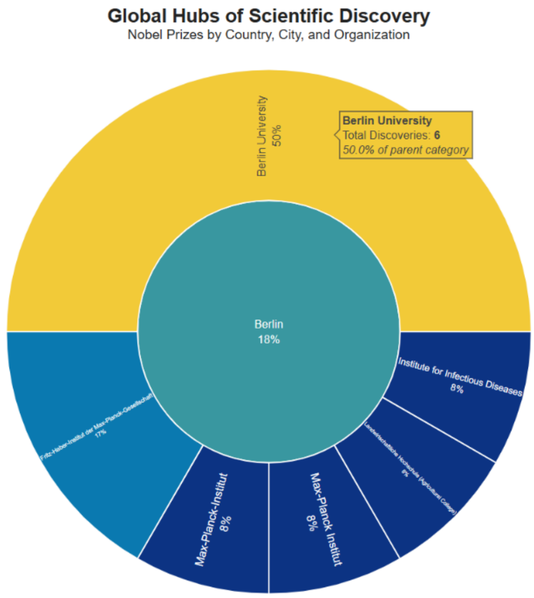
Graph:



Interactive graph link: [Global Hubs of Scientific Discovery]

Data for Germany:





The sunburst chart provides a multi-layered view of the distribution of Nobel Prizes, allowing us to seamlessly drill down from countries to cities and finally to individual institutions, making complex hierarchical relationships easy to understand at a glance.

What stands out is the high concentration of awards in just a few countries—like we saw before—where cities like Cambridge and New York dominate, a trend likely attributable to long-term investments in research funding, world-class universities, and strong academic networks.

Overall, this visualization is extremely insightful, since it not only highlights the world’s leading parts but also illustrates how scientific excellence is geographically concentrated, thereby providing insights into how the environment and infrastructure shape innovation.

Analysis of the Age distribution of Laureates

Now that we have examined the geographical and institutional aspects of Nobel laureates, let us turn to a more personal characteristic: the age at which laureates receive their awards.

This perspective allows us to better understand when scientific breakthroughs typically occur in various fields and time periods.

An analysis of the age distribution reveals patterns regarding professional experience, career paths, and the evolving nature of groundbreaking research.

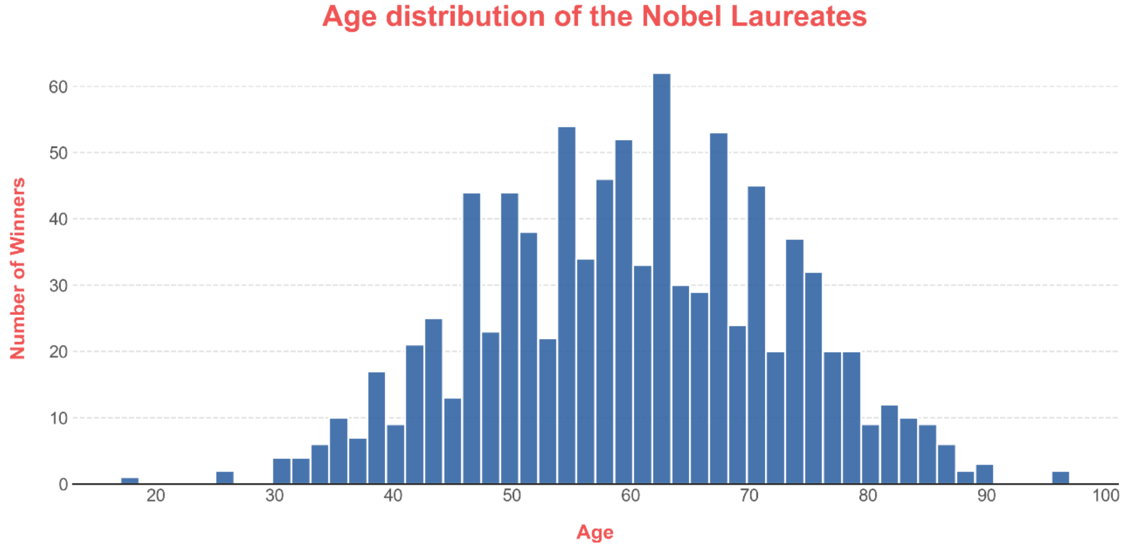
Creating Histogram Plot:

To illustrate this distribution, we will use Seaborn's histplot, which is well-suited for visualizing the distribution of numerical data across intervals.

In this method, age groups are grouped into categories and their frequencies are displayed, making it easy to identify common age groups, trends, and unusual patterns such as early or late career breakthroughs.

```
sns.set_theme(style="white", context="notebook")

plt.figure(figsize=(12, 6), dpi=300)

ax = sns.histplot(
    data=df_data,
    x='winning_age',
    bins=50,
    color="#3465a4",
    edgecolor="white",
    linewidth=1.5,
    alpha=0.9
)

plt.title('Age distribution of the Nobel Laureates',
          fontsize=22, pad=20, fontweight='bold', color='#F05454')
plt.xlabel('Age', fontsize=15, labelpad=15, fontweight='bold', color='#F05454')
plt.ylabel('Number of Winners', fontsize=15,fontweight='bold', labelpad=15,
color='#F05454')

ax.grid(axis='y', color='#dddddd', linestyle='--', alpha=0.8)
ax.tick_params(axis='both', which='major', labelsize=13, colors='#555555')
ax.set_axisbelow(True)

sns.despine(left=True, bottom=False)
plt.tight_layout()

plt.show()
```

Graph:



The histogram shows that Nobel laureates typically receive their awards between their late 50s and mid-60s, with a distinct peak in their early 60s, suggesting that major scientific recognition often comes only after decades of persistent work.

This pattern likely reflects the time it takes for discoveries to be confirmed and recognized by the scientific community, as well as the Nobel Committee's tendency to award prizes years after the original work—sometimes even decades later.

Interestingly, the relatively small number of very young laureates highlights how rare breakthroughs are early in a career, which supports the view that groundbreaking research is generally the result of many years of experience, collaboration, and accumulated expertise.

Analysis of Age Trends over Time

In this section, we are going to go beyond the general age distribution and examine how the age of Nobel laureates has changed over time. In this way, we can examine whether academic recognition tends to occur earlier or later in researchers' careers across different historical periods.

By analyzing these trends, we can gain insights into how the pace of discovery, validation, and recognition in science has changed over the course of more than a century.

Understanding LOWESS Smoothing

To better capture the underlying trend in the data, we use LOWESS (Locally Weighted Scatterplot Smoothing).

Instead of fitting a single straight line, LOWESS generates a smooth curve that better follows the data, making it easier to identify gradual changes and nonlinear patterns over time.

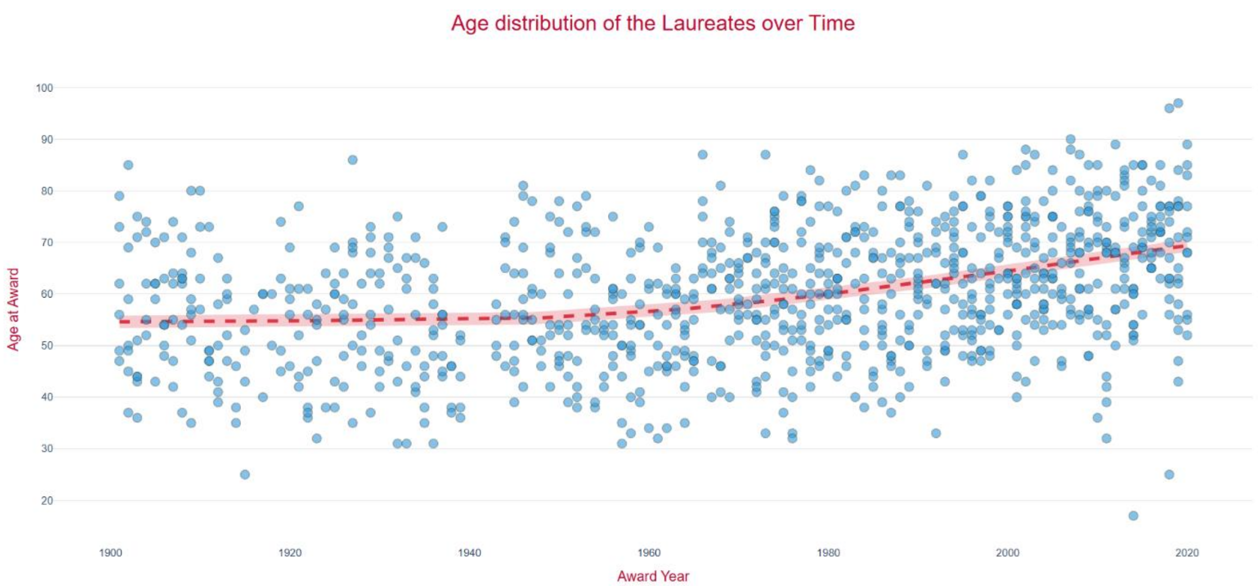
Creating the Scatter Graph with Regression (Trend) line

code:

```
fig_px = px.scatter(
    df_data,
    x='year',
    y='winning_age',
    trendline='lowess',
    hover_name='full_name',
    hover_data={'full_name':False,'category':True,'year':True,'winning_age':
True},
    opacity=0.6,
    title='Age distribution of the Laureates over Time',
    labels={'year': 'Award Year', 'winning_age': 'Age at Award', 'category':
'Prize Category'},
    template='plotly_white'
)

# We are going extract the traces (scatter and trendline) and design them.
scatter_trace = fig_px.data[0]
trendline_trace = fig_px.data[1]

# scatter points
scatter_trace.marker.size = 10
scatter_trace.marker.color = '#3498DB'
scatter_trace.marker.line.width = 0.8
scatter_trace.marker.line.color = 'DarkSlateGrey'

# Trendlines
trendline_trace.line.color = '#DC3545'
trendline_trace.line.dash = 'dash'
trendline_trace.line.width = 4

# we are going to create shadows effect for the trendlines
shadow_trace = go.Scatter(
    x=trendline_trace.x,
```

```
    y=trendline_trace.y,
    mode='lines',
    line=dict(color='rgba(220, 53, 69, 0.25)', width=14),
    hoverinfo='skip',
    showlegend=False
)

# applying the customization.
fig = go.Figure(
    data=[shadow_trace, trendline_trace, scatter_trace],
    layout=fig_px.layout
)

# Layout Design
fig.update_layout(
    font_family="Inter, Helvetica, Arial, sans-serif",
    title_font_size=24, title_font_color="#BF092F", title_x=0.5,
    xaxis=dict(showgrid=False, zeroline=False, title_font_size=16,
title_font_color="#BF092F"),
    yaxis=dict(showgrid=True, gridcolor='#E5E7EB', zeroline=False,
title_font_size=16, title_font_color="#BF092F"),
    plot_bgcolor='rgba(0,0,0,0)',
    hoverlabel=dict(bgcolor="white", font_size=14, font_family="Inter, Helvetica,
Arial, sans-serif")
)


fig.show()
```

Graph:



Interactive Graph link: [Age distribution of the Laureates over Time]

- Each individual entry can be observed with hovering option.

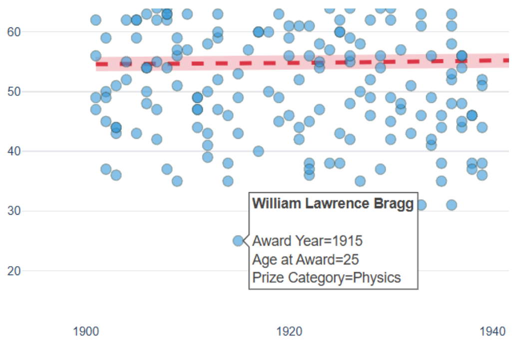
The scatter plot, strengthened with a LOWESS trend line, shows a clear increase in the average age of Nobel laureates over time: from the mid-50s at the beginning of the 20th century to around the late 60s in recent years. Over the past century, the average age of Nobel laureates has increased by at least ten years.

This suggests that significant scientific recognition tends to come later in life, likely due to longer research cycles, the increasing complexity of modern science, and the ever-widening gap between discovery and recognition—studies have shown that this gap has widened considerably over the past century.

The interactive hover feature enhances the analysis by allowing us to examine individual cases (such as recent outliers or historically significant award winners) in greater detail, making the visualization not only informative but also highly engaging and encouraging a deeper exploration.

Identifying Exceptional Outlier in the Graph

While most Nobel laureates do not receive their awards until later in life, the visualization shows a notable outlier at the beginning of the 20th century.



This entry represents a scientist who achieved groundbreaking success at an unusually young age, thus standing in stark contrast to the general trend.

By examining such exceptions, we gain deeper insights into the rare cases in which significant scientific breakthroughs are achieved early in a researcher’s career.

When we hover over that point, we see that this point stands for a Physician called Willim Lawrence Bragg, and he received his award in 1915 when he was 25 years old.

William Lawrence Bragg

A particularly striking example is William Lawrence Bragg, who won the Nobel Prize in Physics in 1915 at the age of just 25, making him the youngest Nobel laureate in the natural sciences to date.

His work in X-ray crystallography, which he conducted together with his father, revolutionized scientists' understanding of atomic structures and laid the groundwork for significant advances in physics and chemistry.

This early achievement is exceptional, as it stands in stark contrast to the general trend of delayed recognition and highlights just how rare it is for such groundbreaking discoveries to be made so early in a scientist's career.

Analysis of Age Distribution Across Categories

In this section, we are going to examine how the ages of Nobel laureates differ across the different prize categories. Instead of viewing age as a single overall distribution, this approach will allow us to compare how career trajectories differ across departments.

In some fields, early theoretical breakthroughs may be rewarded quickly, while in others, recognition comes only after decades of work and as their significance grows.

When we break down age groups by category, we can identify significant differences in how and when peak performance is achieved in various fields.

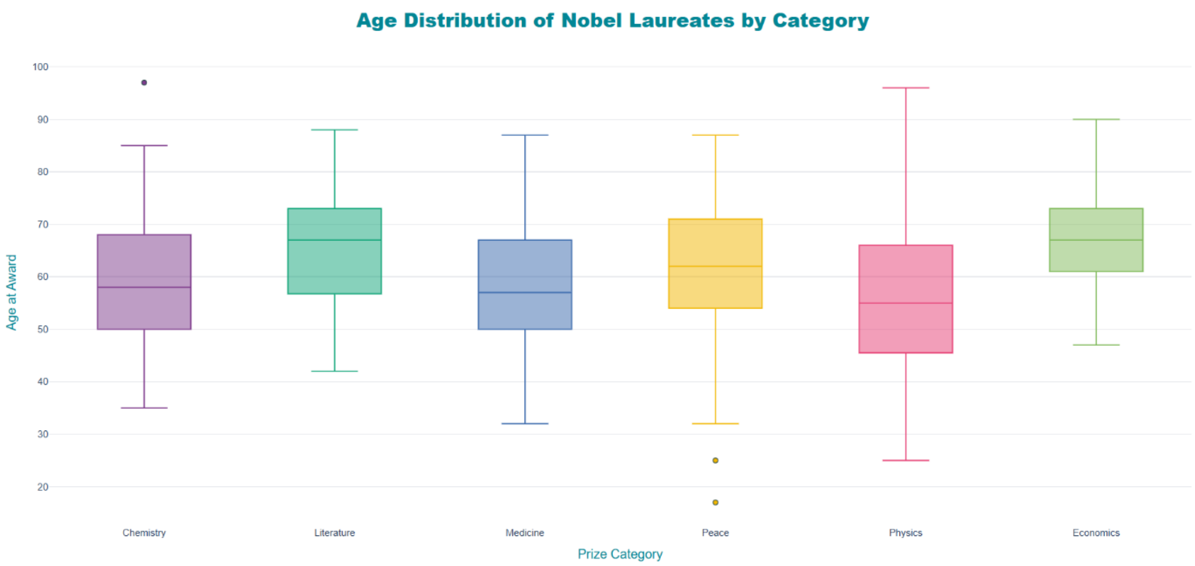
Creating box plot graph:

code:

```
fig = px.box(
    df_data,
    x="category",
    y="winning_age",
    color="category",
    title="Age Distribution of Nobel Laureates by Category",
    labels={"category": "Prize Category", "winning_age": "Age at Award"},
    color_discrete_sequence=px.colors.qualitative.Bold,
```

```
    template="plotly_white"
)

fig.update_layout(
    font_family="Inter, Helvetica, Arial, sans-serif",
    title_font_size=24, title_font_color="#008695", title_font_family="Arial
Black",
    title_x=0.5, # Center the title
    showlegend=False, # No need to show legend
    xaxis=dict(showgrid=False, zeroline=False, title_font_size=16,
tickfont=dict(size=12), title_font_color="#008695"),
    yaxis=dict(showgrid=True, gridcolor='#E5E7EB', zeroline=False,
title_font_size=16, title_font_color="#008695"),
    plot_bgcolor='rgba(0,0,0,0)',
    margin=dict(t=80, b=40, l=40, r=40)
)
fig.update_traces(
    marker=dict(size=6, opacity=1, line=dict(width=1, color='DarkSlateGrey')),
    line=dict(width=1.5)
)

fig.show()
```

Box Plot:



Interactive Graph Link : [Age Distribution of Nobel Laureates by Category]

The box plot clearly shows how age distributions differ across the various Nobel Prize categories: Physics has the longest whiskers, indicating the widest gap between younger and older laureates, while Economics has the highest median age, making it the category with the oldest laureates on average.

In contrast, physics tends to have the youngest laureates on average, which may be due to the fact that theoretical breakthroughs are often achieved early in a scientist's career, whereas fields such as economics and literature generally require longer periods of intellectual development and recognition.

One particularly interesting finding is that, although economics is a relatively young category (introduced in 1969), it includes some of the oldest laureates—likely because it often takes decades for contributions in this field to be validated and gain worldwide recognition, which confirms the general trend that recognition in more interpretive or cumulative disciplines tends to come later.

Analysis of Laureates' Winning Age Trends by Category Over Time

In this section, we expand our analysis by examining how the age at which Nobel laureates receive their awards has changed over time across the various prize categories.

Instead of looking at age trends in general terms, this approach allows us to identify how individual departments differ in terms of career progression and recognition over time.

By tracking these changes over decades, we can better understand whether certain fields are moving toward earlier or later recognition, and what this reveals about the nature of progress in each field.

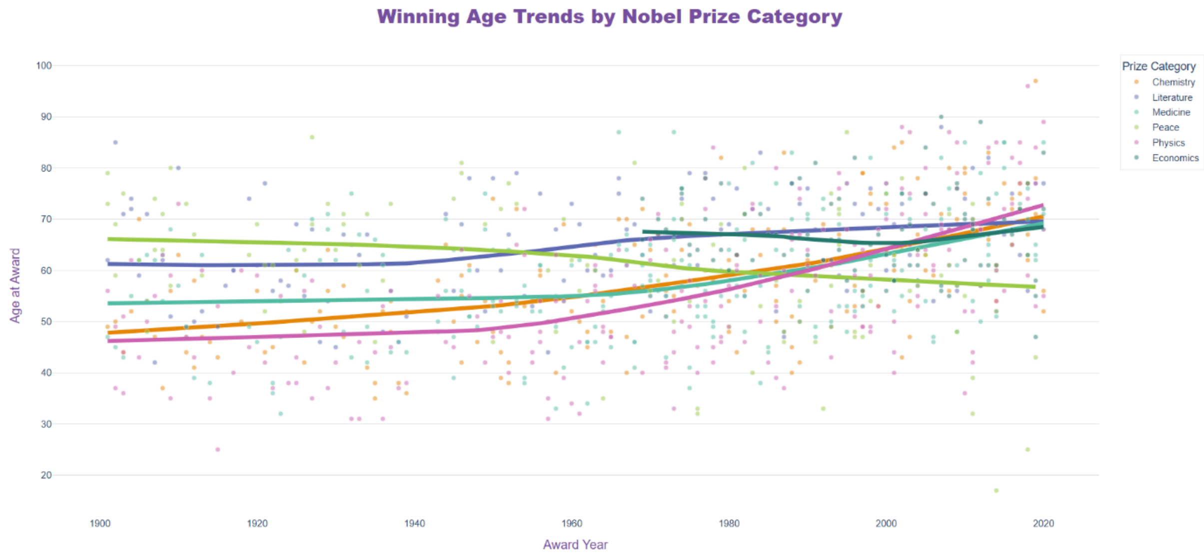
Creating Linear Model Plot :

code:

```
fig = px.scatter(
    df_data,
    x='year',
    y='winning_age',
```

```
    color='category',
    trendline='lowess',
    hover_name='full_name',
    hover_data={ 'category': False, 'year': True, 'winning_age': True},
    title='Winning Age Trends by Nobel Prize Category',
    labels={'year': 'Award Year', 'winning_age': 'Age at Award', 'category':
'Prize Category'},
    color_discrete_sequence=px.colors.qualitative.Vivid,
    template='plotly_white'
)
# Designing for the trend lines
fig.update_traces(line=dict(width=5), selector=dict(mode='lines'))

# Designing for the scatter plots.
fig.update_traces(
    marker=dict(size=6, opacity=0.5, line=dict(width=0.5, color='white')),
    selector=dict(mode='markers')
)

# Layout Arrangements
fig.update_layout(
    font_family="Inter, Helvetica, Arial, sans-serif",
    title_font_family="Arial", title_font_size=24, title_font_color="#764e9e",
    title_x=0.5, # Centers the title
    xaxis=dict(showgrid=False, title_font_size=16, title_font_color="#764e9e"),
    yaxis=dict(showgrid=True, gridcolor='#E5E7EB', zeroline=False,
               title_font_size=16,  title_font_color="#764e9e"),
    # legend layout:
    legend=dict(
        title_font_family="Inter, Helvetica, Arial, sans-serif",
        font=dict(size=12),
        yanchor="top", xanchor="left",
        y=1, x=1.02, # Locating the legend outside the graph
        bgcolor="rgba(255, 255, 255, 0.9)",
        bordercolor="#E5E7EB", borderwidth=1 ),
    plot_bgcolor='rgba(0,0,0,0)',
    margin=dict(t=80, b=40, l=40, r=120), # right margin to fit the legend safely
    hoverlabel=dict(bgcolor="white", font_size=13)
)

fig.show()
```

Graph:



Interactive graph link : [Winning Age Trends by Nobel Prize Category]

The chart shows that the average age of laureates has gradually increased over time in most categories, while categories such as the Peace have shown a slight decline. This suggests that modern scientific achievements often require longer periods of research, whereas in fields such as peace, the effects may become apparent earlier.

The most compelling aspect of this analysis, however, is the visualization itself: By using an interactive Plotly scatter plot with trend lines, we can dynamically explore over 800 individual data points, making it possible to filter out categories, and examine specific award winners in more detail.

This approach significantly increases analytical depth, transforming a static observation into a discovery tool that simultaneously reveals both general patterns and detailed insights.

Conclusion:

As part of this project, we analyzed data on Nobel laureates from various perspectives, examining age distribution, historical trends, and differences across categories, while also identifying notable outliers such as the youngest and oldest laureates. We analyzed gender disparities, highlighted influential figures such as Marie Curie and William Lawrence Bragg, and examined how scientific achievements are distributed across countries, cities, and institutions to paint a comprehensive picture of how cutting-edge research has evolved over time.

Beyond the results, this project demonstrates the consistent application of data science methods, ranging from data cleaning and feature engineering to advanced visualization and exploratory analysis. By combining tools such as pandas for data processing and Plotly and Seaborn for interactive and statistical visualizations, we transformed raw data into meaningful insights using techniques such as grouping, aggregation, smoothing, and hierarchical modeling. The use of interactive elements, custom graphics, and well-designed visualizations not only improves clarity but also demonstrates the ability to handle data both technically and creatively—an essential skill in practical data science work.

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